Response to the Editor and Reviewer Comments

2025-RA-20071.R2

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Content, Source, and Position: The Impacts of Platform-Specific Internal Generative AI on Online Q&A Communities

We sincerely thank the Senior Editor, the Associate Editor, and the reviewers for the opportunity to revise the paper. This revision is grounded in the detailed guidance we received from the Senior Editor, through both email correspondence and offline face-to-face discussion. Following that guidance, we have fundamentally strengthened the paper in the following areas:

- We rebuilt the theoretical development under a single overarching framework, casting the human–AI encounter as a competitive episode. Drawing on competition dynamics theory (Chen et al., 2007; Chen, 2009) and Kahneman's fast-and-slow theory (Evans & Stanovich, 2013; Kahneman, 2003, 2011), we now explain how a human member appraises the AI rival and how that appraisal governs whether to contribute and how much effort to invest.
- We simplified and focused the story on how an internal AI answer competes with human contributors. The effect of an AI answer now depends on its profile: the effect of AI label alone draws human contribution, while the high perceived coverage induced by AI length deters it, and the two components jointly determine the direction of the average overall effect of internal GenAI in our setting.
- We realigned the empirical analysis with this framework so that each mechanism test maps directly to a theoretical argument, and we narrowed the scope by downplaying elements that are less connected to our current story, removing the net-likes results and the position effect from the manuscript.
- We sharpened the paper's positioning relative to prior work by showing how a single framework reconciles our findings with Su et al. (2025), through whether the internal AI is read as a complement to or a substitute for human contributors.

We have also carefully addressed all additional concerns raised by the reviewers. Below, we present our detailed itemized responses to each comment. We hope the review panel finds the paper significantly improved from the previous version.

Deleted: We deeply appreciate the opportunity for major revision and are committed to addressing all concerns with the rigor. Following the editors' and reviewers' guidance, we have fundamentally strengthened the paper in the following areas:
We have simplified the story and focuse on how AI answer compete with, crowd out, and trigger the innovation of human answerers. We downtone the discussion on likes and position, which we found are not unique to AI.
built an overarching framework based on the xxx theory and yyy theory to direct the development of the paper
We have strenghtern the mechanism

To facilitate the review process, we have prepared a comprehensive Online Appendix (https://researchbox.org/5255&PEER_REVIEW_passcode=UFACJZ) containing additional analyses, robustness checks, and supplementary materials that address the review panel's comments. While the main manuscript presents our core findings and primary analyses, the Online Appendix provides complementary evidence that further supports the robustness of our results.

#	Summary	Concerns	To do
R2-8	Clarification		Done
R2-14	Analysis	Statistical test	Done

R2-17	Analysis	Regress # to 1st	Done
R2-21			
R2-22			Done
R2-23			Done
R2-24			Done
R3-9	Correct		Done
R3-10	Analysis		Done
R3-11	Clarficatin		Done
SE-3; R1-5	Difference from Su 2023		
SE-5	Three-path	• ~~Incentive: IGAI alter value of human and expected reward~~ • ~~Authority: Evidence AI is higher authority~~ • Competition between human and AI	
SE-1; R1-7; R2-1; R2-2; R3-7	Over arching theory	R1: dual process theory not OK	
SE-2; AE-2; R1-12; R2-3; R2-15; R3-1	Content Effect	R1: What is length R2: AI is worse than human R2: AI is not higher quality R3: length is not quality AE: show AI has higher quality	Better than some
SE-2; AE-3; R1-13; R17; R3-2	Position Effect	R1: Why care about absolute 1st R2: effect neglectable R2: regress # with high_1st R3: why 1st	ok
SE-2; AE-4; R1-6; R1-14; R2-18; R3-3 R2-4	Source Effect	R1: Controdict with the ban policy R2: GAI authority is not AI authority R2: Residual can not be used R3: Residual can not be used	Yes. Agree
R3-4	Attentin diversion	Hard to conclude	

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R2-26			
R1-8; R2-6; R3-5	Quality degrade	R1: not distingurisable R2: recogion is absolute R3: can not rule out	Merge NO
R1-9; R3-6	elevated expectations	R1: Contridict with the ban policy R3: answer quality as moderating	
R2-25			
R3-6	Analysis		
R1-16	Analysis	Comment sentiment	
R3-1	Analysis	Causal mediation analysis	
R2-26	Discussion	XXXX	
R1-10; R2-10; R1-17; R2-7; R3-8	data	AI rank 1st? Sample restriction	
R2-12	identification	Random effect	
SE-4	Conversation with Lit.		
R1-1	Motivation	Argue internal GenAI is different from external GenAI	
R1-3; R1-4	Objective	Is external GenAI removed?	two AI have different effect

SE Comments

#*SE-0

To evaluate your work, I worked with the same Associate Editor and review panel to evaulate your work. As you are well-aware, this panel is familiar with both your topic and your empirical approach. I would like to begin by expressing my appreciation for the care and effort the Associate Editor and the reviewers invested in evaluating this revision. The Associate Editor's report in particular reflects a careful reading of the manuscript, the revision document, and the reviewers' comments, as well as thoughtful reflection on how best to proceed.

I also want to acknowledge the Associate Editor's reluctance to recommend that the manuscript move forward in the review process. Their report reflects a sincere effort to balance the promise of the research with the review team's concerns. I appreciate that deliberation and the care taken in reaching their recommendation.

The review team's assessment remains mixed, with most reviewers continuing to express significant concerns about the manuscript. However, after carefully reviewing the revision, the Associate Editor's report, and the reviewers' comments, I found myself reluctant to reject the paper. The author team has addressed many of the panel's empirical concerns. However, they have not made sufficient progress in addressing the panel's theoretical concerns about the paper. Given the progress on one dimension, but not the other, this presents a quandary - and the more prudent editorial path is likely to reject the paper.

However, I believe there is a narrow path forward for the paper if the theoretical issues identified in the review package can be addressed convincingly.

Accordingly, I am willing to offer one additional revision opportunity.

I want to be clear: this revision must do more than resolve empirical concerns; it must squarely address the theoretical issues raised by the Associate Editor and the reviewers. While the empirical work in the paper is promising, the manuscript cannot move forward in the review process unless the theoretical explanation of the observed effects becomes substantially clearer and more convincing.

Response

We sincerely thank the review panel for the recognition of our progress on the empirical dimensions and the opprntunity to revise the paper. We are fully committed to addressing these theoretical concerns substantively and demonstrating a convincing explanation of the mechanisms underlying our findings in this revision.

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#SE-1

The Associate Editor has done an excellent job summarizing the review team's concerns. Rather than repeating their detailed discussion, I will highlight the issues I believe are essential to address in the revision.

First, the paper must establish a clearer and stronger theoretical foundation.

The central issue identified by the Associate Editor and all three reviewers is that the *manuscript currently documents empirical patterns without providing a sufficiently convincing theoretical explanation of what mechanisms are generating those patterns.*

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As the Associate Editor notes, the paper reasonably demonstrates that the treatment effects are robust and that the empirical patterns are unlikely to be artifacts of model specification or data issues. However, documenting the pattern alone is not sufficient. The review team needs to be convinced that the manuscript provides a plausible and theoretically grounded explanation of the mechanisms underlying those patterns.

Currently, the manuscript relies on three proposed mechanisms: content effect, source effect, and position effect. Reviewer 2 questions whether the selection of these mechanisms is theoretically justified relative to other mechanisms discussed in the literature. Reviewer 3 similarly notes that the three mechanisms appear to be introduced largely in isolation rather than as part of a coherent theoretical argument. The Associate Editor raises a related concern: the manuscript lacks an overarching theoretical framework explaining why these mechanisms should operate together.

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In the next revision, the theoretical argument must be reconstructed to present a coherent explanation of how internal generative AI changes participation incentives and evaluation processes in online knowledge communities.

This requires more than refining the discussion of the three effects. The paper must articulate:

• the underlying theoretical problem the study addresses

• the mechanism or mechanisms through which internal GenAI alters contributor behavior

• how the proposed mechanisms follow from existing theory

Put simply, the paper must move from describing empirical regularities to explaining the behavioral processes that generate those regularities.

Please note that, because MISQ prioritizes theoretical contributions, this concern must be substantively addressed in this revision. I view this as the "paper-breaker" - if you can't address the concern about the theoretical foundation, then the paper can not advance to another round of review.

Response

We treat this as the central concern of the revision. The underlying theoretical problem the study addresses is how the arrival of an internal AI answer changes a human member's two decisions in an online knowledge community: whether to contribute at all, and how much effort to invest. In this revision we rebuild the theoretical foundation so that the paper explains the behavioral processes that generate the empirical patterns, rather than only documenting the patterns. We made two changes, described below.

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The xxx theory

First, a reorganization of how the theory is presented. We have rebuilt the theoretical development based on the email communication and offline face-to-face instruction with SE. It begins from: **1) a competitive encounter perspective** motivated by competition dynamics theory (Chen et al., 2007; Chen, 2009), which explains how the member appraises the AI rival; and **2) a single overarching account Kahneman's fast-and-slow theory** (Evans & Stanovich, 2013; Kahneman, 2003, 2011), which explains which signal can affect each decision and when that signal becomes available..

Grounded in new framework, we now claim that the effect of an AI answer depends on the **AI profile**. The AI label alone signals that a competitor has entered the question. In a community that already views AI-generated content with suspicion, users treat this competitor as a weak one, and the presence of a weak competitor motivates human contribution (H1). The AI length works in the opposite direction. When the AI answer is long and appears comprehensive, potential answerers judge that little of value remains to add, and their contribution falls (H2). The positive prediction is thus derived from theory **before analysis and stated in the hypotheses themselves**, which makes the ordering between theory and findings explicit.

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Our results confirm this framework. The AI label alone effect on supply is positive and significant (0.188, $p<0.05$), the interaction with AI answer length is negative (-0.046, $p<0.001$), and the average total effect turns negative only because AI answers in our setting carry substantial content, which is identically the total deterrence effect in the original manuscript.

Second, a deliberate narrowing of scope to keep the story focused. The removal is a choice about narrative focus and length, not a retraction of the earlier logic. We removed three elements from the main text. (1) All results that use net likes as the dependent variable, together with every recognition mechanism built on them, are removed. (2) The position effect is removed from the main text. (3) In the response letter, we still engage fully with the reviewer's earlier concerns about these removed elements, in order to show that the original reasoning was sound.

#SE-2

Second, the empirical tests of the mechanisms must more directly support the theoretical explanation.

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Many of the reviewers' concerns stem from the fact that the empirical analyses intended to test the proposed mechanisms do not provide clear evidence for them.

For example:

Reviewer 1 and Reviewer 3 both question whether the analysis in Section 6.3.1.1 provides convincing evidence for the proposed content effect, noting that answer length alone is not a reliable measure of answer quality.

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Reviewer 2 and Reviewer 3 raise concerns about the operationalization of the position effect, particularly the use of the "first-position seeker" construct and its connection to the underlying theoretical logic.

All three reviewers question the inference of the source effect, noting that attributing residual treatment effects to an AI authority mechanism is not justified unless other plausible mechanisms have been ruled out.

These issues suggest that the current analyses of the mechanism do not yet provide convincing empirical support for the theoretical claims. In the revision, the mechanism tests should be reconsidered to more directly correspond to the theoretical arguments being advanced.

Please note, that this may require your team to develop a novel theoretical explanation. Given you results counter Su et al. (2023), you may need to develop a novel theoretical account for your findings.

Response

We are grateful for this guidance and for the opportunity to discuss it further, both in the review letter and in our conversation. Following your suggestion, we have reconsidered the mechanism tests so that they correspond more directly to the theoretical arguments we advance, and we have reorganized the theoretical account around the fast and slow processing of AI cues. Below we respond point by point and indicate how each response maps to the corresponding revision. For the underlying reviewer concerns, we also provide detailed replies in R1-x, R2-x, and R3-3.

On the content effect and the concern that answer length alone is not a reliable measure of quality (raised by Reviewer 1 and Reviewer 3).

We looked into this carefully and now treat length in three steps.

First, we clarify that answer length measures coverage rather than overall quality. In our original framing, we imprecisely characterized length as a quality proxy. We now position it specifically as a measure of answer coverage, which captures the breadth and thoroughness of the answer. This is conceptually distinct from the dimensions captured by our LLM-based measures. As detailed in Appendix B, the LLM assessment focuses on objective quality dimensions such as accuracy, clarity, and relevance, which evaluate how well an answer addresses what it chooses to cover rather than how much ground it covers. This distinction is consistent with the mediation pattern we observe. Potential human answerers are deterred primarily by the perceived high coverage of the AI answer, a surface cue that can be inferred from length at a glance, rather than by fine-grained quality, which becomes available only through more effortful evaluation.

Second, we report the three validation tests we conduct for the LLM-based quality measure. Failed AI answers (for example, "I'm sorry, I can't provide a detailed answer") received significantly lower quality ratings than substantive answers ($p < 0.001$). Human answers accepted by questioners received significantly higher ratings than non-accepted answers ($p < 0.05$). Answers with above-median net likes received significantly higher ratings than those below the median ($p < 0.001$). These results support the validity of the measure.

Third, we examined what answer length actually captures. Length can reflect perceived quality, or it can reflect perceived coverage, meaning the impression that an answer spans many aspects of the problem. To separate these two readings, we added a second content variable, the number of bullet points in the answer. GenAI answers in this setting are typically well formatted and well structured, and an answer that enumerates several points tends to convey broad coverage, which motivates a perception of comprehensiveness. If length were operating mainly through perceived coverage, its effect should track this coverage variable rather than a direct quality signal. Our results indicate that the role of length operates largely through perceived coverage rather than through perceived quality. We report this analysis in the mechanism section.

On the position effect and the operationalization of the "first-position seeker" construct (raised by Reviewer 2 and Reviewer 3).

The reviewers' concerns converge on the same underlying issue with this approach. The analysis rested on an implicit assumption that first-position seekers care about being absolutely first rather than first among human contributors. This assumption followed from our competitive framing, in which the human answerer and the AI compete for the same scarce resource, namely the limited attention and recognition of viewers, so that an existing AI answer reduces the value of the first human answer for a first-position seeker. On reflection, we agree that testing this mechanism adequately in our setting would require data on how answerers perceive and value position, which our current dataset does not provide. We therefore cannot adequately separate a position-based account from the other mechanisms that the reviewers note could generate the same empirical pattern. For this reason, and to keep the theoretical account focused on the fast and slow processing of AI cues in a way that aligns with our updated framework, we have removed the position effect from the manuscript in this revision.

On the inference of the source effect and the attribution of residual treatment effects (raised by all three reviewers).

We accept this concern in full. A residual cannot be cleanly attributed to a source-based mechanism, both because alternative channels that our controls do not capture may drive the reduction in human answers, and because any remainder may reflect measurement error or an incomplete operationalization of the other channels. We therefore no longer rest any claim on a residual. Under the revised framework, the AI label effect and the AI content effect are each theorized and estimated on their own terms, rather than one being inferred as what remains after the other is accounted for. We elaborate this estimation strategy in our detailed reply to R3-3.

Taken together, these revisions move the mechanism tests closer to the theoretical arguments they are meant to support, and they organize the account around the fast and slow processing of AI cues rather than around a residual-based decomposition.

#SE-3

> *Third, the manuscript must more clearly situate its findings relative to prior work.*
>
> *Several reviewers raise concerns about the paper's positioning relative to existing literature, particularly Su et al. (2023).*
>
> *As Reviewer 1 notes, the current explanation for the divergence between your findings and prior work is not yet convincing. The revision should clarify why the mechanisms proposed in your paper would lead to different outcomes in your setting and what theoretical insight emerges from this comparison.*
>
> *This clarification is important for establishing the paper's contribution.*

Response

Following the suggestion of the review panel, we have reorganized our differentiation from Su et al. (2025) around a single idea: what separates the two settings is whether the internal AI serves as a *complement* to human contributors or as a *substitute* that competes with them. Two observable factors jointly determine which case holds. The first is the community's evaluation culture (Context). The second is the mode of AI deployment (Implementation Mode). Table R1-5 positions the two studies along both factors, together with their designs, findings, and mechanisms. Below we address the reviewer's three points, mapping each to the corresponding row of the table, and then show how a single framework reconciles the two sets of findings.

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Table R1-5. Positioning the Current Study in Internal GenAI Empirical Research: A Comparison with Su et al. (2025)

Dimension	**Su et al. (2025)**	**Our Study (2026)**
Community	Quora	SegmentFault
Community Domain (Context)	General, conversation-oriented	Technical, correctness-oriented
How AI Answer Is Provided (Implementation Mode)	Added in batch to **existing questions** that already hold human answers	**Randomly provided to newly posted questions**, as the first and often the only answer
Research Design	Natural Exp. + Experiment	Randomized Exp.
Unit of Analysis	Question-Week	Question
Key Findings	↑ Human answers	↓ Human answers; ↑ Answer quality

Factor 1. Community evaluation culture (Community Domain row; addresses Point 1).

We agree that the "general-interest versus technical" framing was overstated, and we have removed the claim that our users are technical experts while Quora's are casual users. Two clarifications sharpen the comparison. First, the updated version of Su et al. (2025) no longer includes the objective-versus-subjective accuracy analysis, so this contrast is no longer part of their design. Second, their expert classification does not track technical expertise: it is based on Quora's Top Writer badge, which "is awarded to contributors based on the quality of their responses," so that "contributors with the Top Writer badge on Quora are identified as human experts while those without the badge are identified as non-experts" (Su et al. 2025). Their experts are strong contributors on a conversation-oriented platform, not specialists solving verifiable technical problems.

We therefore reframe this difference as one of evaluation culture rather than user composition (Burtch et al. 2024). Quora primarily serves everyday, conversational topics; questions such as "What did Marilyn Monroe carry in her coffin?" and "What is the best small business to start in Gambia?" illustrate this orientation,[1] and Oliveira et al. (2018) document that Quora exhibits significantly more "conversation ingredients" than Stack Overflow. SegmentFault, like Stack Overflow, operates under an efficiency-oriented philosophy and seeks a single correct answer to each question.

This cultural difference maps directly onto whether the AI is read as a complement or a competitor, which in turn sets the direction of the effect on contribution. In a conversation-oriented culture, an answer is one contribution among many, and the AI is read as a **complement**. It functions as a catalyst rather than a rival: the AI and human contributors draw on different strengths and are not competing on the same dimension, so a contributor who sees an AI answer feels that the discussion has been re-energized rather than closed off. The effect on contribution is therefore positive. In a correctness-oriented culture, where each question calls for one right answer, the AI is read as a direct **competitor**. Here, once contributors see AI answer content, a strong perception of coverage signals that the question is already resolved and deters contribution. The effect on contribution is therefore negative. We re-elaborate this in Section

Research Gap and Section Conclusion.

Factor 2. Mode of AI deployment (How AI Answer Is Provided row; addresses Points 2 and 3).

We accept that the phrase “three months” was imprecise, because the three-month figure is the observation window on each side of the policy launch. The substantive point, however, holds. In Su et al. (2025), the questions were created before the panel began, and the AI answers were added only at the policy launch. Their design keeps “only those questions that were created before the start of our panel data collection (i.e., November 2022),” and the AI answers appear when “AI answers are presented on treated question pages” at the launch (Su et al. 2025). The AI therefore enters questions that were posted earlier and had already accumulated human answers. We have rewritten the passage to state this precisely rather than to attach a specific number to the gap between question posting and AI answer arrival.

This distinction speaks directly to the reviewer’s third point. We agree that timing per se is not the mechanism; what matters is whether, at the moment of decision, the AI is read as one more voice alongside existing human answers or as the standing answer from competitor that the human contributor must compete with. In Su et al. (2025), the AI joins a thread that already contains human answers on a platform where multiple perspectives coexist, so it enters as an additional voice which may catalyze the conversation and is read as a **complement**. The effect on contribution is therefore positive. In our setting, the AI is the **competitor** that competes with humans for limited attention and recognition from viewers, for the same goal of providing a correct solution. Therefore, perception of comprehension after seeing AI answer content may deter the human contribution.

The two factors thus push in the same direction within each study. The conversational culture and the batch-to-existing-threads deployment both render the AI a complement in Su et al. (2025), while the correctness-oriented culture and the first-answer-to-new-questions deployment both render it a competitor in ours. We re-elaborate this in Section *Research Gap* and Section *Conclusion*.

Reconciling the two sets of findings within one framework.

Our framework accommodates both sets of results without appeal to a separate theory. Grounded in new framework, **we now claim that the effect of an AI answer depends on the AI profile.** Our results confirm this framework. The **AI label alone** effect on supply is positive and significant (0.188, $p<0.05$), the interaction with **AI answer length** is negative (-0.046, $p<0.001$), and the average total effect turns negative only because AI answers in our setting carry substantial content, which is identical to the total deterrence effect in the original manuscript.

The two components behave differently depending on whether the AI is a complement or a substitute. When the AI is a complement, whether because of the community’s conversational culture or because it enters an established thread, the label effect is positive, since the AI acts as a catalyst that revives discussion. The length deterrence effect, in contrast, is largely absent. Because the AI and human contributors draw on different strengths, a longer and more complete AI answer does not lead a contributor to conclude that there is nothing left to add. In this case the AIlabel effect dominates and the net effect on contribution is positive, which matches the results of Su et al. (2025). When the AI is a competitor, as in our technical setting where it is the first answer to a new question judged on correctness, a longer AI answer signals that the question is already well covered and deters contribution. The content effect dominates and the net effect is negative. Both findings follow from the same two components, and the direction of the net effect depends on whether the AI is read as a complement or a substitute.

Reference

Oliveira, N., Muller, M., Andrade, N., & Reinecke, K. (2018). The exchange in StackExchange: Divergences between Stack Overflow and its culturally diverse participants. *Proceedings of the ACM on Human-Computer Interaction, 2*(CSCW), 1-22.

[1] Examples from actual Quora content. See: https://www.quora.com/What-did-Marilyn-Monroe-carry-in-her-coffin and https://www.quora.com/What-is-the-best-small-business-to-start-in-Gambia

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#SE-4

Fourth, the manuscript should clarify the relationship between theory and empirical findings.

Reviewer 2 raises a concern that the theoretical development may have been shaped after the empirical results were observed. Whether or not that characterization is accurate, the manuscript should clarify the logical ordering between theory and empirical analysis.

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In particular, the revision should explicitly acknowledge alternative theoretical predictions that exist in the literature. Some prior work suggests that AI-generated answers could increase rather than decrease human participation. Addressing these alternative possibilities directly will strengthen the credibility of the theoretical argument.

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Response

Following the SE's guidance, we have placed our theoretical development in explicit conversation with prior studies, clarified the logical ordering between theory and analysis, and directly acknowledged the alternative predictions in the literature. We summarize the changes below and indicate where they appear in the revised theoretical development section.

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Organizing the alternative mechanisms within one framework (R2-1). Learning, crowding out, anchoring, and cognitive offloading are all relevant to this setting, and in this revision we do not treat them as competing alternatives to be excluded. Instead, we adopt an overarching framework, Kahneman's dual-process theory, that organizes these forces and specifies which mode of processing governs each of the answerer's decisions (Evans & Stanovich, 2013; Kahneman, 2003, 2011). The suggested mechanisms map directly onto this framework rather than sitting outside it: **Learning** corresponds to the deliberate, System 2 engagement in H6, where an answerer who reads a high-quality AI answer raises the quality of their own contribution. **Cognitive offloading and the crowding out of effort** operate through the law of least effort that governs the fast stage, and enter the framework through H2, where a comprehensive AI answer changes the perceived cost and benefit of contributing. **Anchoring** is

Formatted: Paragraph, Space Before: 0 pt, After: 0 pt, No bullets or numbering the explicit core of H4: the weak-competitor judgment formed from the bare AI label acts as an anchor on subsequent effort, so that the member expects even a fairly modest answer to be enough to come out ahead (Kahneman, 2011).

Adopting a single overarching framework, rather than a list of separate mechanisms, allows us to explain the two decisions an answerer actually faces, whether to contribute and how much effort to invest, within one coherent account. Following this reorganization, the "when and where the AI appears" dimension is no longer part of the framework; the position effect has been removed from the main text to keep the theoretical story focused on the fast and slow processing of AI cues.

Logical ordering of theory and analysis (R2-2). The observed net reduction in human answers is the sum of two forces that are both theorized in advance: the positive AI label alone effect in H1 and the negative AI content effect in H2, with the content-driven force dominating in our setting. The positive prediction is thus derived from theory before analysis and is stated in the hypotheses themselves rather than inferred after the fact, which makes the ordering between theory and findings explicit.

Reconcile the prior work suggesting that AI may increase participation. Prior work suggesting that AI-generated answers could increase rather than decrease human participation is accommodated within our framework, without appeal to a separate theory. We claim that the effect of an AI answer depends on the AI profile. In our data the AI label effect on supply is positive and significant (0.188, $p < 0.05$), its interaction with AI answer length is negative (-0.046, $p < 0.001$), and the average total effect turns negative only because AI answers in our setting carry substantial content. This total effect is identical to the overall deterrence effect reported in the original manuscript.

The two components behave differently depending on whether the AI acts as a complement or a substitute. **When the AI is a complement**, whether because of a community's conversational culture or because it enters an established thread, the label effect is positive, as the AI acts as a catalyst that revives discussion, while the length deterrence effect is largely absent, since the AI and human contributors draw on different strengths and a longer, more complete AI answer does not lead a contributor to conclude that nothing remains to add. Here the AI label effect dominates and the net effect on contribution is positive, which is consistent with the results of Su et al. (2025). **When the AI is a competitor**, as in our technical setting where it is the first answer to a new question judged on correctness, a longer AI answer signals that the question is already well covered and deters contribution; the AI content effect dominates and the net effect is negative. Both patterns follow from the same two components, with the direction of the net effect depending on whether the AI is read as a complement or a substitute.

#SE-5

So what to do? Given the strength and breadth of objections to your work?

The review package suggests several directions to strengthen the paper's theoretical contribution. While the choice of approach ultimately rests with the authors, the following directions strike me as particularly promising.

One possible direction is to frame the paper more explicitly around participation incentives in online knowledge communities. In this framing, the introduction of internal GenAI could

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A second possibility is to develop a stronger theoretical account of authority and credibility signals in human–AI interaction. Several reviewers note that the paper currently assumes that users attribute authority to AI-generated answers, but this assumption requires stronger theoretical justification in the context of generative AI systems.

A third direction is to frame the study more broadly as examining competition and complementarity between AI-generated and human-generated knowledge in digital communities. Such a framing could help explain when AI discourages human participation and when it might instead stimulate additional contributions.

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These suggestions are not intended to prescribe a specific solution. Rather, they illustrate the type of theoretical narrative that will be necessary for the paper to make a convincing contribution.

Response

We are grateful to the SE for laying out these directions. After working through them, we have reorganized the paper around the third one, **framing the study as an examination of competition and complementarity** between AI-generated and human-generated knowledge in a digital community. The revised theoretical development now opens by casting the human-AI encounter as a competitive episode. We summarize the reorganization below and indicate where it appears.

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We treat the encounter as follows. A member posts a question, the platform attaches an answer from its own AI system carrying a label that identifies the author as a machine, and a human member who arrives later must decide two things: whether to write an answer at all, and, if so, how much effort to put into it. We develop a theory of how the member sizes up the machine rival and how that assessment shapes what the member does next.

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We give three reasons for adopting the competition lens, and we state them so that the logical basis of the framing is explicit rather than assumed. First, the lens fits the situation the platform has created, since the AI answer sits in the same thread and on the same screen as any human answer that follows, so a later human answer has to be worth reading in the presence of the AI answer. Second, the lens tells us where to look, because research on competitive interaction holds that how an actor responds to a rival depends on how the actor appraises that rival, and that this appraisal is built from observable features such as who the rival is, how capable it appears, and how much effort it appears to have put in (Chen et al., 2007; Chen, 2009). This is also where the member's response to the AI label enters as an appraisal of an observable cue rather than as an assumption. Third, the competition lens generates richer predictions than a substitution account. Prior work that treats generative AI as a substitute for human labor establishes that these systems can perform much of the routine work human contributors once did, but that account can predict only decline, because if the AI answer does work the member would otherwise have done, the member has less left to do and writes less. It contains no mechanism by which an AI answer could lead a member to write more, or to write better.

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Within the competition framework, the AI label effect and the AI content effect are distinct components whose relative weight determines whether the net effect on contribution is negative or positive, so the same account explains both when AI discourages human participation and when it can stimulate additional or higher-effort contributions. The full development, including the mapping of learning, crowding out, and anchoring onto this framework, is presented in the revised theoretical development section (see also our response to SE-4).

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AE Comments

#*AE-0

> *As previously, the three reviewers once again did great jobs in objectively and constructively evaluating the revised paper. They commented on authors' responses to the previous suggestions and pointed out remaining and new issues. The three reviewers' comments are once again highly consistent. Unfortunately, instead of converging positively, reviewers remain concerned and critical in general. The collective recommendation from the reviewer panel is still negative, with only one reviewer indicating a further revision may be hopeful yet expressing more serious concerns than in the last round.*
>
> *I very carefully read the revised paper, the authors' revision notes, and the referee reports. I commend the authors' effort in a round of diligent revision, resulting in a much-improved paper. They addressed many of the issues highlighted in the SE and AE reports from last round. As a result, the concerns regarding the randomization have largely been eliminated; the main treatment effects have been shown stable when covariates are progressively added to the model; many data integrity questions have been clarified; the difference from existing literature has been discussed in detail.*

Response

We sincerely thank the Associate Editor for the thorough and careful evaluation of our revised manuscript, as well as the recognition of improvements in empirical analysis and literature review. We appreciate the review panel's further comments and we are committed to addressing these issues. We hope the review panel finds the paper significantly improved.

In this round of revision, we benefited greatly from extensive communication with the **Senior Editor, including both email correspondence and offline face-to-face discussion**. Following SE's **detailed and actionable guidance**, we have made the following major changes. **First**, we substantially refined the theoretical framework along the competition direction suggested by the Senior Editor, establishing the theoretical hypotheses under a coherent and overarching logic. **Second**, we realigned our empirical results with this framework, which involved both adding results that speak more directly to the theory and removing those that were less central to it. **Third**, we sharpened the positioning of our contribution by more explicitly articulating how our study differs from prior work. We elaborate on each of these changes in our point-by-point responses below.

#AE-1

> *Nonetheless, I have to agree with the reviewers that key concerns linger, namely, regarding mechanism analysis and theorization, which in essence go hand in hand.*
>
> *I believe the authors managed to establish that the main treatment effects are stable and robust, utilizing the unique opportunity of the randomized experiment of this focal platform. However, the review team feels that merely documenting such patterns, which in many ways contradict prior findings in the literature, is not enough to create meaningful new knowledge about what exactly is at play that leads to such phenomena. They need to be*

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convinced of a plausible explanation of the underlying mechanism, supported by solid theories---ideally from a sound overarching theoretical framework. This paper appears to fall short in this regard unfortunately.

Empirically, the analysis exploring the mechanism is somewhat flawed and hence unconvincing, as all three reviewers unanimously pointed out and I summarize in detail below. It then loops back to the issue of lacking a convincing theoretical framework, as all three reviewers complain. R2 questioned the approach here appears to be HARKing (i.e., hypothesizing after results are known). Such inadequacy further leads to the doubts on the discrepancy between the main effects found here and those reported in the literature (e.g., see R1's comments on Su et al, 2023).

Response

We are grateful to the Associate Editor and the review team for directing our attention to theorization and mechanism. We took these concerns seriously and, following the email communication and the offline face-to-face instruction with the SE, undertook a systematic reconstruction of the paper. Our revision responds in two parts. The first part rebuilds the theoretical development under a single overarching framework and aligns the empirical results with it. The second part explains how our main effects reconcile with prior findings, in particular Su et al. (2025).

=================== Part 1: Theory and Empirical Results ===================

First, a reorganization of how the theory is presented. We have rebuilt the theoretical development based on the email communication and offline face-to-face instruction with SE. It begins from: **1) a competitive encounter perspective** motivated by competition dynamics theory (Chen et al., 2007; Chen, 2009), which explains how the member appraises the AI rival; and **2) a single overarching account Kahneman's fast-and-slow theory** (Evans & Stanovich, 2013; Kahneman, 2003, 2011), which explains which signal can affect each decision and when that signal becomes available..

Grounded in new framework, we now claim that the effect of an AI answer depends on the AI profile. The AI label alone signals that a competitor has entered the question. In a community that already views AI-generated content with suspicion, users treat this competitor as a weak one, and the presence of a weak competitor motivates human contribution (H1). The AI length works in the opposite direction. When the AI answer is long and appears comprehensive, potential answerers judge that little of value remains to add, and their contribution falls (H2). The positive prediction is thus derived from theory before analysis and is stated in the hypotheses themselves, not inferred after the fact.

Our results confirm this framework. The AI label alone effect on supply is positive and significant (0.188, $p<0.05$), the interaction with AI answer length is negative (-0.046, $p<0.001$), and the average total effect turns negative only because AI answers in our setting carry substantial content, which is identically the total deterrence effect in the original manuscript.

Our mechanism tests now examine the arguments developed in the theoretical section directly. Each test maps to a specific argument in the theoretical development, so the mechanism analysis and the theory are now aligned one to one. For every comment that questioned the

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earlier mechanism analysis, we have provided a point-by-point response. The details appear in our responses to the individual reviewers.

Second, a deliberate narrowing of scope to keep the story focused. The removal is a choice about narrative focus and length, not a retraction of the earlier logic. We removed three elements from the main text. (1) All results that use net likes as the dependent variable, together with every recognition mechanism built on them, are removed. (2) The position effect is removed from the main text. (3) In the response letter, we still engage fully with the reviewer's earlier concerns about these removed elements, in order to show that the original reasoning was sound.

We hope you find this way of theoretical development under a coherent overarching theoretical framework clearer and satisfactory.

=============== Part 2: Reconciles our findings with Su et al. (2025) ===============

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Regarding the discrepancy with Su et al. (2025), we clarify that our approach differs in its fundamental assumptions. We propose a new theoretical framework that reconciles our findings with theirs by identifying the specific conditions under which these divergent results converge.

We have reorganized our differentiation from Su et al. (2025) around a single idea: what separates the two settings is whether the internal AI is a complement to human contributors or a substitute that competes with them. Two observable factors jointly determine which case holds. The first is the community's evaluation culture (Context). The second is the mode of AI deployment (Implementation Mode). Table AE-1 positions the two studies along both factors, together with their designs, findings, and mechanisms. Below we address the reviewer's three points, map each to the corresponding row of the table, and then show how a single framework reconciles the two sets of findings.

Table AE-1. Positioning the Current Study in Internal GenAI Empirical Research: A Comparison with Su et al. (2025)

Dimension	**Su et al. (2025)**	**Our Study (2026)**
Community	Quora	SegmentFault
Community Domain (Context)	General, conversation-oriented	Technical, correctness-oriented
How AI Answer Is Provided (Implementation Mode)	Added in batch to **existing questions** that already hold human answers	**Randomly provided to newly posted questions**, as the first and often the only answer
Research Design	Natural Exp. + Experiment	Randomized Exp.
Unit of Analysis	Question-Week	Question
Key Findings	↑ Human answers	↓ Human answers; ↑ Answer quality

Factor 1. Community evaluation culture (Community Domain row; addresses Point 1).

We agree that the "general-interest versus technical" framing was overstated, and we have removed the claim that our users are technical experts while Quora's are casual users. Two clarifications sharpen the comparison. First, the updated version of Su et al. (2025) no longer includes the objective-versus-subjective accuracy analysis, so this contrast is no longer part of their design. Second, their expert classification does not track technical expertise: it is based on Quora's Top Writer badge, which "is awarded to contributors based on the quality of their responses," so that "contributors with the Top Writer badge on Quora are identified as human experts while those without the badge are identified as non-experts" (Su et al. 2025). Their experts are strong contributors on a conversation-oriented platform, not specialists solving verifiable technical problems.

We therefore reframe this difference as one of evaluation culture rather than user composition (Burtch et al. 2024). Quora primarily serves everyday, conversational topics; questions such as "What did Marilyn Monroe carry in her coffin?" and "What is the best small business to start in Gambia?" illustrate this orientation,[1] and Oliveira et al. (2018) document that Quora exhibits significantly more "conversation ingredients" than Stack Overflow. SegmentFault, like Stack Overflow, operates under an efficiency-oriented philosophy and seeks a single correct answer to each question.

This cultural difference maps directly onto whether the AI is read as a complement or a competitor, which in turn sets the direction of the effect on contribution. In a conversation-oriented culture, an answer is one contribution among many, and the AI is read as a **complement**. It functions as a catalyst rather than a rival: the AI and human contributors draw on different strengths and are not competing on the same dimension, so a contributor who sees an AI answer feels that the discussion has been re-energized rather than closed off. The effect on contribution is therefore positive. In a correctness-oriented culture, where each question calls for one right answer, the AI is read as a direct **competitor**. Here, once contributors see AI answer content, a strong perception of comprehensiveness signals that the question is already resolved and deters contribution. The effect on contribution is therefore negative. We re-elaborate this in Section Research Gap and Section Conclusion.

Factor 2. Mode of AI deployment (How AI Answer Is Provided row; addresses Points 2 and 3).

We accept that the phrase "three months" was imprecise, because the three-month figure is the observation window on each side of the policy launch. The substantive point, however, holds. In Su et al. (2025), the questions were created before the panel began, and the AI answers were added only at the policy launch. Their design keeps "only those questions that were created before the start of our panel data collection (i.e., November 2022)," and the AI answers appear when "AI answers are presented on treated question pages" at the launch (Su et al. 2025). The AI therefore enters questions that were posted earlier and had already accumulated human answers. We have rewritten the passage to state this precisely rather than to attach a specific number to the gap between question posting and AI answer arrival.

This distinction speaks directly to the reviewer's third point. We agree that timing per se is not the mechanism; what matters is whether, at the moment of decision, the AI is read as one more voice alongside existing human answers or as the standing answer from competitor that the human contributor must compete with. In Su et al. (2025), the AI joins a thread that already contains human answers on a platform where multiple perspectives coexist, so it enters as an additional voice which may catalyze the conversation and is read as a **complement**. The effect on contribution is therefore positive. In our setting, the AI is the **competitor** that competes with humans for limited attention and recognition from viewers, for the same goal of providing a correct solution. Therefore, perception of comprehension after seeing AI answer content may deter the human contribution.

The two factors thus push in the same direction within each study. The conversational culture and the batch-to-existing-threads deployment both render the AI a complement in Su et al. (2025), while the correctness-oriented culture and the first-answer-to-new-questions deployment both render it a competitor in ours. We re-elaborate this in Section *Research Gap* and Section *Conclusion*.

Reconciling the two sets of findings within one framework.

Our framework accommodates both sets of results without appeal to a separate theory. Grounded in new framework, **we now claim that the effect of an AI answer depends on the AI profile.** Our results confirm this framework. The **AI label alone** effect on supply is positive and significant (0.188, $p<0.05$), the interaction with **AI answer length** is negative (-0.046, $p<0.001$), and the average total effect turns negative only because AI answers in our setting carry substantial content, which is identical to the total deterrence effect in the original manuscript.

The two components behave differently depending on whether the AI is a complement or a substitute. When the AI is a complement, whether because of the community's conversational culture or because it enters an established thread, the label effect is positive, since the AI acts as a catalyst that revives discussion. The length deterrence effect, in contrast, is largely absent. Because the AI and human contributors draw on different strengths, a longer and more complete AI answer does not lead a contributor to conclude that there is nothing left to add. In this case the AIlabel effect dominates and the net effect on contribution is positive, which matches the results of Su et al. (2025). When the AI is a competitor, as in our technical setting where it is the first answer to a new question judged on correctness, a longer AI answer signals that the question is already well covered and deters contribution. The content effect dominates and the net effect is negative. Both findings follow from the same two components, and the direction of the net effect depends on whether the AI is read as a complement or a substitute.

Reference

Oliveira, N., Muller, M., Andrade, N., & Reinecke, K. (2018). The exchange in StackExchange: Divergences between Stack Overflow and its culturally diverse participants. *Proceedings of the ACM on Human-Computer Interaction, 2*(CSCW), 1-22.

[1] Examples from actual Quora content. See: https://www.quora.com/What-did-Marilyn-Monroe-carry-in-her-coffin and https://www.quora.com/What-is-the-best-small-business-to-start-in-Gambia

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#AE-2 (content effect)

> *Mechanism driving reduction in human answers*
>
> *(a) Content effect*
>
> *As all three reviewers pointed out and I also agree, the fact that AI answers have higher quality than human answers has not been well established in the analysis of 6.3.1.1. The authors only show the direct testing result that AI answers have greater length than human answers, which is neither surprising nor supportive of the claim that AI answers have higher content quality than human answers.*
>
> *As R1 further observed, the focal platform has some guidelines against external GenAI tools, and users of this platform expressed substantial skepticism and dissatisfaction toward AI-generated answers in general. It further raises the question whether users indeed view AI-generated answers as trustworthy and having higher authority, which is used as the foundation of the whole mechanism analysis throughout the paper.*

Response

We thank the AE and the reviewers for prompting us to look more carefully into this issue. Following the comment, we revisited two related concerns: first, whether the content of AI answers is strong enough to affect human contribution; and second, whether users in this technical Q&A domain indeed regard AI answers as trustworthy and authoritative, which the comment notes is the premise underlying the mechanism analysis. We address these two concerns in turn below.

Part 1. Whether and how the AI content is strong enough to affect human contribution.

Why the AI answer is strong enough to matter. We do not claim that AI dominates all humans. The platform's human contributors form a certain distribution of ability, and while AI's quality may not exceed that of the best human, it certainly exceeds that of the weaker contributors; otherwise the platform would not have deployed the system at all. This is sufficient for a "crowding-out" effect: the AI answer will deter the segment of potential rookie contributors who think AI is not weak in their mind, because they think the answers the AI has already supplied what they could add. This is the basic argument for our content effect.

Why perceived AI quality deters contribution: perceived coverage as a fast signal. Beyond this logic, we also theorize the perceptual channel through which AI content operates, since what deters a potential contributor is the *perceived* rather than the objectively verified quality of the AI answer. Consistent with the fast-and-slow account, potential contributors do not systematically evaluate the AI answer before deciding whether to contribute; instead they rely on readily available surface cues. AI answers are characteristically long and heavily structured into bullet points, and this length and structure serve as a heuristic (fast) signal of high coverage. An answer that *looks* thorough raises the potential contributor's perception that the information need has already been met, which lowers the perceived gap that a new human answer could fill and thereby deters contribution. This is why the content effect operates through perceived comprehension, and it is reflected in our empirical test of AI answer length and AI answer

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quality in the results section.

Part 2. How users perceive the AI label, and its role in the mechanism.

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The first part clarifies how the content of the AI answer bears on contribution. Turning to the second concern, the skepticism and dissatisfaction the reviewer observed in this domain led us to reconsider the role we had assigned to the AI label itself.

On reexamination, and consistent with the user reactions the reviewer points to, we now read the label differently. In this technical Q&A domain, the bare AI label triggers skepticism rather than trust. As developed in the theoretical development part (H1), seeing the label activates a routine assessment that, through the availability heuristic (Tversky & Kahneman, 1973), retrieves the prevailing domain-specific impression that AI answers are unreliable on genuine technical problems. The answerer therefore appraises the AI as a weak competitor and is drawn toward contribution rather than away from it (H1).

The skepticism and dissatisfaction the reviewer identified in the user comments under the announcement are the sentiment this mechanism relies on, and we now use them as supporting evidence for H1. Because the AI label alone does not by itself deter contribution in this domain, the deterrence we observe operates largely through the answerer's evaluation of the AI length rather than through the label alone. The reviewer's reading of the context is thus incorporated directly into the revised theory.

Taken together, both parts of our reexamination point to a mechanism reading that fits this domain more closely: the deterrence works through the perceived coverage of the AI length rather than through trust in the AI label, and we have revised the theory and H1 accordingly.

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#AE-3 (position effect)

(b) Position effect

All three reviewers have issues with the approach surrounding "first-position seekers" for exploring the position effect. See R1's comments on page 4, R2's point 5 on the last page, and R3' comments in (b).

Response

We thank the AE and all three reviewers for their comments on the position effect. Reading R1's comment on page 4, R2's point 5, and R3's comment in (b) together, we recognize that the reviewers' concerns converge on the same underlying issue with our "first-position seekers" approach. In particular, the analysis rested on an implicit assumption that first-position seekers care about being absolutely first rather than first among human contributors

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The assumption itself followed from our competitive framing, in which the human answerer and the AI compete for the same scarce resource, namely the limited attention and recognition of viewers, so that an existing AI answer reduces the value of the first human answer for a first-position seeker. On reflection, however, we agree that testing this mechanism adequately in our setting would require data on how answerers perceive and value position, which our current

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dataset does not provide. We therefore cannot adequately separate a position-based account from the other mechanisms that the reviewers note could generate the same empirical pattern.

For this reason, and to keep the theoretical story focused on the fast and slow processing of AI cues in a way that aligns more closely with our updated framework, we have removed the position effect from the manuscript in this revision. Importantly, the substance of the reviewers' concern is not lost. The consideration that R2 emphasizes, that the first position in the treatment group is occupied by an AI answer and that AI and human answers are processed differently, is now internalized in the H1, which argues the effect of the AI label alone. What was previously developed as a separate position mechanism is now embedded in the development of the AI label effect, whose theorizing already presumes that the AI answer is the first mover occupying the dominant first position and is therefore processed as a fast, heuristic signal. In this sense the framework has not changed in substance so much as in how it is presented, with the interaction between position and source now folded into a single, more focused mechanism rather than treated separately.

We have addressed the more specific points raised by each reviewer in our replies to R1-13, R2-5, and R3-2, and we do not repeat those details here. We are grateful that these comments helped us clarify the boundary of what our focused framework now claims.

#AE-4 (source effect)

> *(c) Source effect*
>
> *As all three reviewers questioned and I feel very much the same way, the analysis leading to the conclusion on the "source effect" of AI-generated answers appears to be the most unsound. Simply attributing all the remaining effects of AI answers on the decrease of human answers other than the (questionable) content and position effects to the "source effect" and interpreting it as a heightened perceived authority users give to AI answers seem too hasty and unjustified. On the very contrary, the large proportion of the unexplained residual effects (i.e., 54%) indicates there are very likely unmodeled alternative mechanism at work. See R1's last point on page 4, R2's comments on R2-20, and R3's comments in (c).*
>
> *It becomes more questionable when considered in connection with the above points in (a) that AI answers may not necessarily have higher content quality and users may hold skepticism toward AI answers. The results in 7.1 that such authority effect does not exist for high-quality human answers further adds to the doubt about whether the perceived authority of AI answers really exists or not. As R2 rightly pointed out, it is unsubstantiated to extrapolate prior findings on users' trust on predictive AI to support the claim on users' trust on generative AI, because the latter may appear less reliable in nature due to issues like hallucination.*

Response

We are grateful for this integrated guidance and for the opportunity to revisit the source effect carefully. We took these concerns seriously and re-examined both how we identified the effect

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and how we presented it. This reconsideration led us to change our approach substantially in the current version, and we describe that change first, since it underlies our responses to the individual reviewers on this issue.

In the previous version, we began from a total AI effect, accounted for a content channel and a position channel, and interpreted the remaining portion as a source effect reflecting heightened perceived authority. We no longer take this approach. In the current version, grounded in a coherent overarching framework, we claim that the effect of an AI answer depends on the AI profile, and we separate two distinct dimensions that an answerer encounters at different moments: the AI label alone, which is visible before any content is read, and the AI content, in particular its length, which becomes relevant only after the answer is examined. Both predictions are derived from theory in advance and estimated directly, rather than inferred from a residual after the fact. This is a change in how we identify and present the mechanisms, not a reversal of our earlier finding that AI presence deters human contribution on average.

On the residual-based attribution. We accept the concern in full. A residual cannot be cleanly attributed to a source-based mechanism, both because alternative channels that our controls do not capture may drive the reduction in human answers, and because any remainder may reflect measurement error or an incomplete operationalization of the other channels. We therefore no longer rest any claim on a residual. Under the revised framework, the AI label alone effect and the AI content effect are each theorized and estimated on their own terms. We elaborate this in our detailed reply on the R3-3.

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On users' skepticism toward AI and why we earlier viewed the source as plausible. Our earlier argument about the AI authority was framed at the level of the overall effect. In the revised analysis we no longer attribute the effect at the overall level; instead we attribute it at the level of the AI label alone. In this technical Q&A domain, seeing the bare label activates a routine assessment that, through the availability heuristic (Tversky & Kahneman, 1973), retrieves the prevailing domain-specific impression that AI answers are unreliable on genuine technical problems. The answerer therefore appraises the AI as a weak competitor and is drawn toward contribution rather than away from it. The skepticism and dissatisfaction expressed in the user comments are precisely the sentiment this account relies on, and we now use them as supporting evidence for this prediction. The deterrence we observe on average operates largely through the answerer's evaluation of the AI content and its length, not through the label alone. We develop this reasoning in our detailed reply on the R1-6.

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On the extrapolation from predictive AI. We accept this as well and have removed the analogy to predictive AI. Findings on authority and credibility in forecasting contexts cannot be transported directly to generative AI, which can fabricate and hallucinate and which users may appraise very differently. The effect of the bare AI label is now explained without any appeal to that literature: it follows from the answerer's domain-specific doubt about the reliability of AI-generated content on the technical problem at hand, which is fully consistent with the fact that generative AI can hallucinate. We set this out in our detailed reply on r2-4.

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On whether the perceived authority of AI answers exists, given that no such effect appears for high-quality human answers. We appreciate this observation and have looked into it carefully. On reflection, the authority attached to an AI answer and the authority attached to a human expert may be different in kind, since the former is machine-related and AI-specific rather than a straightforward analogue of human expertise, so the absence of an authority effect for high-

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quality human answers does not by itself settle whether an AI-specific effect exists. To keep the theoretical framework close to the results and to present the more convincing evidence that are more tightly linked to the theoretical framework, we have removed this part of the analysis from the current version and no longer build any claim on the perceived authority of AI answers.

We hope you find this way of organizing the theory and evidence, under a single coherent framework, clearer and more satisfactory.

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#AE-5 (viewer recognition mechanism)

2. Mechanism driving reduction in viewer recognition

Reviewers are also unsold on the mechanism analysis leading to the reduction in viewer recognition. Both R2 (point 6 on the last page) and R3 (point d) questioned the attention diversion mechanism. R3 (point e) questioned the quality degradation mechanism. Both R1 (page 3) and R3 (point f) questioned the elevated expectation mechanism. As R1 pointed out, elevated expectation would make sense only if AI-generated answers are indeed of higher quality and regarded more trustworthy by users, which is an unsubstantiated assumption by itself and loops back to the discussion in (a) and (c) above.

Response

We thank the AE for consolidating the reviewers' concerns on the mechanisms underlying the reduction in viewer recognition. We carefully investigated each of these concerns across the three mechanisms, and we summarize our examination below, with detailed responses provided under the corresponding reviewer comments.

On the attention diversion mechanism (R2 point 6; R3 point d), the reviewers noted that the null result on commenting activity does not necessarily rule out attention diversion, and that the coefficient magnitude for comments appears similar to that for net likes. Following this, we clarify that likes and comments share a necessary precondition, namely that the viewer must read and attend to the human answer, so if attention were diverted then commenting should decline as well, which we do not observe; we also note that the two outcomes are on different scales (net likes are not logged while comments are logged) and capture different things (recognition versus attention-driven activity), so a similar coefficient magnitude does not indicate a similar effect. Taken together, the pattern is more consistent with a reduction in recognition than with a diversion of attention, and the full logic is provided in our responses to R2-26 and R3-4.

On the quality degradation mechanism (R3 point e), the reviewer noted that similar expertise among answerers does not rule out the possibility that the same users exert less effort when an AI answer is present, which could lower human answer quality. We examined both pathways through which quality degradation could operate, a compositional shift toward lower-expertise users and reduced effort by the same users, and we find no evidence of a compositional shift (expertise, measured by prior accepted answers within the question's tag area, does not decline) and no evidence of reduced answer quality (LLM quality ratings show no association, and answer length is if anything slightly higher). Since quality is determined by effort conditional on an unchanged composition of answerers, the absence of a quality decline suggests effort is likely not reduced, though we acknowledge this is indirect evidence as we do not have a direct measure of effort, as detailed in our response to R3-5.

On the elevated expectations mechanism (R1 page 3; R3 point f), R1 noted that the mechanism relies on users perceiving AI answers as superior and norm-setting, which requires clearer justification, and R3 suggested using AI answer quality as a moderator. In our current view this assumption is captured in H2 of our updated framework, where potential answerers are deterred by the perceived high coverage of the AI answer inferred from its length through attribute substitution (Kahneman 2011; Kahneman & Frederick 2002), developed from the reviewer's idea of AI answers as superior; because control-group questions never receive an AI answer, a direct AI-quality moderator is not feasible, so we instead use the quality of the first answer as a continuous moderator, and the interaction is negative and significant ($\beta_1 = -0.037$, $p < 0.05$), consistent with elevated evaluative standards, as detailed in our responses to R1-9 and R3-6.

As stated in our opening remark, to make the paper more focused and to ensure the results align more strongly with our updated theoretical framework, we removed all netlike-related results in the manuscript. But we retain the underlying logic here in response to the reviewer's concern.

#AE-6 (data intergrity)

In addition, reviewers also pointed out some remaining data integrity issues, for example, R1's first comment on page 4, R2's comments on R2-7, R2-8, and R-16.

Response

We thank the AE for consolidating these points. We reviewed each of them carefully and confirm that they concern reporting and typographical matters rather than the underlying analyses. We have addressed all of them: we corrected the typographical error noted in R1's comment (R1-11, R1-18) and conducted a comprehensive proofreading of the entire manuscript to ensure no similar inconsistencies remain, and we took concrete steps to improve transparency and corrected the typos in our earlier response as noted in R2-7 and R2-8, as well as the point raised in R2-16.

The detailed corrections are provided in our responses to R1-11, R1-18, R2-7, and R2-8.

#AE-7

As much as I hate to see papers get rejected in the second round, I also hate to have the review process dragging on without a clear direction to pursue and a promising end in sight. Given that pretty much the entire review team remain concerned and negative about issues as fundamental as discussed above in the second round, after reflecting on reviewers' comments and deliberating for a long while, I feel it is probably in the best interest of both the authors and the review team to call it quits at this point.

This being said, I would certainly encourage the authors not to give up on this research, especially the valuable data from this unique randomized field experiment. They just need to spend more time to think about and explore deeply what exactly are the mechanisms at work

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Response

We deeply appreciate the Associate Editor's thoughtful deliberation and candid assessment of our manuscript's challenges, as well as the encouragement to continue pursuing this research despite the recommendation. We are grateful for the recognition that the randomized field experiment represents valuable data with potential for meaningful contribution. The Associate Editor's guidance to think more deeply about the mechanisms at work and how they reconcile with existing theories provides a clear direction for our revision efforts. We are committed to investing the time and intellectual effort necessary to develop a more convincing theoretical account that transforms our empirical observations into meaningful new knowledge.

Reviewer 1 Comments

#*R1-0

> *Thank you for revising and resubmitting the paper. Although I appreciate the authors' efforts to address my earlier comments, I find that the manuscript still has several major issues that limit its theoretical and practical contribution. I outline below a set of comments that I hope will help the authors further improve the rigor, clarity, and overall contribution of the work.*

Response

We sincerely thank Reviewer 1 for the continued engagement with our manuscript and the recognition of our revision efforts. We appreciate the reviewer's commitment to helping us improve the rigor, clarity, and overall contribution of this work. The detailed comments provided offer valuable guidance for addressing the remaining major issues. We are grateful for the constructive spirit of the review and will carefully address each point raised.

In this round of revision, we benefited greatly from extensive communication with the **Senior Editor, including both email correspondence and offline face-to-face discussion**. Following SE's **detailed and actionable guidance**, we have made the following major changes. **First**, we substantially refined the theoretical framework along the competition direction suggested by the Senior Editor, establishing the theoretical hypotheses under a coherent and overarching logic. **Second**, we realigned our empirical results with this framework, which involved both adding results that speak more directly to the theory and removing those that were less central to it. **Third**, we sharpened the positioning of our contribution by more explicitly articulating how our study differs from prior work. We elaborate on each of these changes in our point-by-point responses below.

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#R1-1 (motivation)

> *– In the opening paragraphs of the introduction, the authors distinguish internal from external GenAI largely from a researcher's perspective (e.g., data observability, attribution challenges, experimental feasibility). While these points explain why the setting is methodologically attractive, they do not sufficiently motivate why the phenomenon itself is theoretically important. What ultimately matters is not that internal GenAI is easier for researchers to study, but that it represents a fundamentally different sociotechnical and governance arrangement from external GenAI. The former involves platform control, explicit labeling, institutional endorsement, and altered legitimacy and accountability. I encourage the authors to reframe the motivation around these substantive differences and use the data-access advantages as supporting justification rather than the primary motivation for the research question.*

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Response

We thank the reviewer for this comment. We agree that the earlier motivation explained why internal GenAI is convenient to study rather than why the phenomenon itself is theoretically important, and that the substantive difference lies in platform control, explicit labeling, institutional endorsement, and the altered legitimacy and accountability of the AI answer. We have reframed the motivation accordingly.

The introduction now opens the internal versus external distinction with a substantive claim rather than a methodological one. The third paragraph states that "internal GenAI is not external GenAI relocated inside the platform. It is a distinct sociotechnical and governance arrangement, and this distinction is what makes the phenomenon theoretically important." We then identify the three features the reviewer points to: the platform controls the AI directly, labels its output explicitly as AI-generated, and places it in the first-answer position. We explain that these features change the legitimacy and accountability of the AI answer, so that the AI answer is no longer a hidden aid to individual answerers but a visible, platform-backed participant that competes with human answers for the limited attention and recognition of viewers.

Deleted: We agree with the reviewer that Internal GenAI represents fundamental differences to platform due its internarity to the organization. Study it can better understand how AI interact with human within the boundary of platform.

Following the reviewer's guidance, we have demoted the data-access advantage to a supporting justification. The fourth paragraph now presents visibility as a consequence of the governance arrangement and a unique empirical advantage rather than as the reason for studying it. Because the platform labels the AI answer and randomly decides which questions receive one, the entry of an AI competitor is a visible and well-defined event, and this is what allows us to isolate an AI effect that stays under private external GenAI use. The data advantage now follows from the substantive difference rather than standing in for it.

Please see the Section Introduction for details.

Deleted: In the revision, we have better articulated the motivation of the paper.

#R1-2

– I appreciate the authors for providing additional contextual details, such as the platform's announcement about the restricted use of external GenAI tools (https://segmentfault.com/a/1190000043031126). I recommend the authors including these details in the paper due to their critical importance.

Response

Thank you for the comment. We have now incorporated the announcement link into the manuscript.

#R1-3 (objective)

– In the response letter, the authors note that SegmentFault instituted policies discouraging or restricting the use of external GenAI tools. Yet, the revised manuscript repeatedly frames the contribution as estimating the incremental effect of internal GenAI in an environment where users have access to external tools like ChatGPT. These two claims appear in tension. If external GenAI use is meaningfully restricted, then the baseline environment is not one of "external GenAI availability," and the interpretation and generalizability of the findings change. The authors should clarify the extent and enforcement of this policy and align the paper's framing and claimed contribution accordingly.

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Response

We thank the reviewer for pressing us to be precise. The apparent tension comes from an ambiguity in our earlier wording, which we have now resolved. The community's restriction targets one specific behavior: posting AI-generated answers on the platform, that is, submitting output copied from an external tool as one's own work. The restriction does not target, and in practice cannot target, a user consulting ChatGPT or learning from it before posting.

This distinction is now stated directly in the introduction. The paragraph reads: "The community in our setting discourages the posting of AI-generated answers, that is, answers that a user copies from an external tool and submits as the user's own work. It does not, and in practice cannot, prevent users from consulting ChatGPT or learning from it before they post." We then define the baseline precisely: "The baseline against which we measure the effect of internal GenAI is therefore one in which external GenAI remains available for private use and continues to divert demand."

The two claims are therefore not in tension. External GenAI availability, understood as private consultation on the supply side and demand diversion on the demand side, remains valid under the restriction. What the platform removes is only the public posting of unattributed external AI output. Our estimate is the effect of adding a visible, platform-integrated internal GenAI answer on top of this baseline, which is now stated as such in the introduction section.

Please see the Section Introduction for details.

Deleted: We appreciate the reviewer's comment. Per the review panel's suggestion in the last round, our paper elaborates that what we identify is the internal GenAI effect given external GenAI's availability. While we emphasize on the community's attitude toward external GenAI to explain one of the reasons they bring Internal GenAI into the platform, it should be noted that it is almost impossible to prevent user from using ChatGPT or learning from ChatGPT before making posts. In this regard, our identified effect is still internal GenAI's effective given the availability of external GenAI (in a way similar as corporate cell phone's impact given personal cellphone).
In the revisition, we have clarified this point.

#R1-4

> *– A related concern is the apparent inconsistency between the platform's restriction of external GenAI and its later deployment of its own internal GenAI. According to the policy link cited in the response letter, the restriction on tools such as ChatGPT is motivated largely by concerns about inaccuracy and hallucination. This raises an important question: why would users view the platform's internal GenAI as sufficiently reliable to warrant prominent first-position answers, when external GenAI is discouraged for reliability reasons? From a user perspective, this asymmetry may undermine trust or generate frustration, potentially discouraging participation for reasons unrelated to the content, source, or position mechanisms proposed in the paper. The authors should clarify how the platform justified this distinction to users and discuss whether governance inconsistency or user resentment could contribute to the observed effects, as well as what this implies for the generalizability of the findings.*

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Response

We thank the reviewer for this important comment, which prompted us to sharpen both the theory and the results. Our revised framework resolves the puzzle the reviewer identifies, and we address the reviewer's three specific requests in turn.

Before responding to individual comments, we describe two changes that run across our responses. Later comments refer back to this note rather than repeating it.

First, a reorganization of how the theory is presented. We have rebuilt the theoretical development based on the email communication and offline face-to-face instruction with SE. It begins from: **1) a competitive encounter perspective** motivated by competition dynamics theory (Chen et al., 2007; Chen, 2009), which explains how the member appraises the AI rival; and **2) a single overarching account Kahneman's fast-and-slow theory** (Evans & Stanovich, 2013; Kahneman, 2003, 2011), which explains which signal can affect each decision and when that signal becomes available..

Grounded in new framework, we now claim that the effect of an AI answer depends on the **AI profile**. The AI label alone signals that a competitor has entered the question. In a community that already views AI-generated content with suspicion, users treat this competitor as a weak one, and the presence of a weak competitor motivates human contribution (H1). The AI length works in the opposite direction. When the AI answer is long and appears comprehensive, potential answerers judge that little of value remains to add, and their contribution falls (H2). The positive prediction is thus derived from theory before analysis and is stated in the hypotheses themselves, not inferred after the fact.

Our results confirm this framework. The AI label alone effect on supply is positive and significant (0.188, $p<0.05$), the interaction with AI answer length is negative (-0.046, $p<0.001$), and the average total effect turns negative only because AI answers in our setting carry substantial content, which is identically the total deterrence effect in the original manuscript.

Second, a deliberate narrowing of scope to keep the story focused. The removal is a choice about narrative focus and length, not a retraction of the earlier logic. We removed three elements from the main text. (1) All results that use net likes as the dependent variable, together with every recognition mechanism built on them, are removed. (2) The position effect is removed from the main text. (3) In the response letter, we still engage fully with the reviewer's earlier concerns about these removed elements, in order to show that the original reasoning was sound.

Then, we respond to the reviewer's concerns point-to-point.

(1) How the platform justified the distinction to users. As with many new technology adoptions, the platform shows a resistance – adoption path. In order to compete with external GenAI, they put forward two instruments: 1) ban external GenAI, 2) make internal GenAI. While the first instrument can not be enforced since it is done by users, the second instrument is under the full control of the platform. Our study is on the second instrument, which is 10 months after the first instrument. The platform says in the announcement of the internal GenAI that this is an experiment that aims to collect user feedback and see whether this instrument can work.

(2) Whether governance inconsistency or user resentment could drive the observed effect. Our updated framework and our research design together rule this out, and in fact the reviewer's intuition about user skepticism is now built into the theory. First, on the design: AI answers are randomly assigned at the question level. Any resentment or distrust a user holds toward the platform's governance is a person-level or platform-level disposition that applies equally to the questions the user encounters with and without an AI answer. Because assignment is random across questions, such sentiment is constant between the treated and control conditions and is differenced out. The only treatment-specific channel is a user's response to seeing a labeled AI answer on a given question, which is exactly what our H1 currently measures. Second, on the direction of that effect: if governance-related resentment or distrust discouraged participation, the AI label alone effect would be positive as illustrated in our current theoretical development. The positive AI label effect shows that users treat the labeled AI as a weak competitor whose presence invites participation. The community's skepticism toward AI, which the reviewer correctly infers from the restriction, is therefore not a confound. It is the very perception that produces the weak-competitor response at the center of our theory. The deterrence we document does not come from the label or from resentment but from the AI length, the perception of coverage that grows with AI answer length.

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Deleted: While one may question if the identified effect would be the same as a platform without the external GenAI exclusion, we argue that internal GenAI and external GenAI have different mechanisms paths to impact the platform (according to both the literature and our study). External GenAI attracts demand, which is immune from the availability of internal GenAI. External GenAI's impact on supply is through human and thus lack of AI identification. Its use may affect content effect but can not affect the position effect and source effect. We elaborate on these points in the discussion.

(3) Implications for generalizability. Two points follow. First, the weak-competitor perception is tied to a community that views AI as a competitor and with suspicion. In a community where people treat AI as a complement rather than a competitor or are more accepting of AI, the AI label effect could be different, and we now state this community context condition in the Discussion. Second, because our results tie the total effect to identifiable features of the AI answer rather than to a fixed average, seeing the effect from an AI profile perspective also travels to future settings in which AI content improves, as we note at the end of the results and conclusions.

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#*R1-5

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– I appreciate the authors' efforts in differentiating from Su et al. (2023). However, I do not find the listed differences convincing. The authors should reconsider or substantially revise their explanation for why their findings diverge from Su et al. (2023).

The claim that Quora is a "general-interest" platform whereas the present study focuses on technical users is overstated. Quora does host IT- and technology-focused communities, and Su et al. explicitly analyze objective versus subjective questions and expert versus non-expert contributors. Thus, differences in user composition alone cannot explain the opposite findings.

In the response letter and revised paper, the authors repeatedly claim that in Su et al. (2023), AI-generated answers arrive "approximately three months after a question is posted." I could not find support for this statement. In Su et al., the three-month period refers to the observation window before and after the policy launch, not to a delay between question posting and AI answer arrival.+

More importantly, even if one were to assume a delay, it is unclear why this would fundamentally alter contributor incentives. Su et al. conduct a controlled online experiment that holds timing constant and still finds positive effects of AI answers. Moreover, human answerers rarely respond immediately after question posting; what matters is whether an AI answer is visible and salient at the time contributors consider answering. As long as the AI answer occupies a prominent position in the answer section, it should affect human contribution decisions regardless of whether it appears immediately or later.

Response

We thank the reviewer for these corrections. We accept them and have reorganized our differentiation from Su et al. (2025) around a single idea: what separates the two settings is whether the internal AI serves as a *complement* to human contributors or as a *substitute* that competes with them. Two observable factors jointly determine which case holds. The first is the community's evaluation culture (Context). The second is the mode of AI deployment (Implementation Mode). Table R1-5 positions the two studies along both factors, together with their designs, findings, and mechanisms. Below we address the reviewer's three points, mapping each to the corresponding row of the table, and then show how a single framework reconciles the two sets of findings.

Table R1-5. Positioning the Current Study in Internal GenAI Empirical Research: A Comparison with Su et al. (2025)

Dimension	**Su et al. (2025)**	**Our Study (2026)**
Community	Quora	SegmentFault
Community Domain (Context)	General, conversation-oriented	Technical, correctness-oriented
How AI Answer Is Provided (Implementation Mode)	Added in batch to **existing questions** that already hold human answers	**Randomly provided to newly posted questions**, as the first and often the only answer
Research Design	Natural Exp. + Experiment	Randomized Exp.
Unit of Analysis	Question-Week	Question
Key Findings	↑ Human answers	↓ Human answers; ↑ Answer quality

Factor 1. Community evaluation culture (Community Domain row; addresses Point 1).

We agree that the "general-interest versus technical" framing was overstated, and we have removed the claim that our users are technical experts while Quora's are casual users. Two clarifications sharpen the comparison. First, the updated version of Su et al. (2025) no longer includes the objective-versus-subjective accuracy analysis, so this contrast is no longer part of their design. Second, their expert classification does not track technical expertise: it is based on Quora's Top Writer badge, which "is awarded to contributors based on the quality of their responses," so that "contributors with the Top Writer badge on Quora are identified as human experts while those without the badge are identified as non-experts" (Su et al. 2025). Their experts are strong contributors on a conversation-oriented platform, not specialists solving verifiable technical problems.

We therefore reframe this difference as one of evaluation culture rather than user composition (Burtch et al. 2024). Quora primarily serves everyday, conversational topics; questions such as "What did Marilyn Monroe carry in her coffin?" and "What is the best small business to start in Gambia?" illustrate this orientation,[1] and Oliveira et al. (2018) document that Quora exhibits significantly more "conversation ingredients" than Stack Overflow. SegmentFault, like Stack Overflow, operates under an efficiency-oriented philosophy and seeks a single correct answer to each question.

This cultural difference maps directly onto whether the AI is read as a complement or a competitor, which in turn sets the direction of the effect on contribution. In a conversation-oriented culture, an answer is one contribution among many, and the AI is read as a **complement**. It functions as a catalyst rather than a rival: the AI and human contributors draw on different strengths and are not competing on the same dimension, so a contributor who sees an AI answer feels that the discussion has been re-energized rather than closed off. The effect on contribution is therefore positive. In a correctness-oriented culture, where each question calls for one right answer, the AI is read as a direct **competitor**. Here, once contributors see AI answer content, a strong perception of coverage signals that the question is already resolved and deters contribution. The effect on contribution is therefore negative. We re-elaborate this in Section Research Gap and Section Conclusion.

Factor 2. Mode of AI deployment (How AI Answer Is Provided row; addresses Points 2 and 3).

We accept that the phrase "three months" was imprecise, because the three-month figure is the observation window on each side of the policy launch. The substantive point, however, holds. In Su et al. (2025), the questions were created before the panel began, and the AI answers were added only at the policy launch. Their design keeps "only those questions that were created before the start of our panel data collection (i.e., November 2022)," and the AI answers appear when "AI answers are presented on treated question pages" at the launch (Su et al. 2025). The AI therefore enters questions that were posted earlier and had already accumulated human answers. We have rewritten the passage to state this precisely rather than to attach a specific number to the gap between question posting and AI answer arrival.

This distinction speaks directly to the reviewer's third point. We agree that timing per se is not the mechanism; what matters is whether, at the moment of decision, the AI is read as one more voice alongside existing human answers or as the standing answer from competitor that the human contributor must compete with. In Su et al. (2025), the AI joins a thread that already contains human answers on a platform where multiple perspectives coexist, so it enters as an additional voice which may catalyze the conversation and is read as a **complement**. The effect on contribution is therefore positive. In our setting, the AI is the **competitor** that competes with humans for limited attention and recognition from viewers, for the same goal of providing a correct solution. Therefore, perception of comprehension after seeing AI answer content may deter the human contribution.

The two factors thus push in the same direction within each study. The conversational culture and the batch-to-existing-threads deployment both render the AI a complement in Su et al. (2025), while the correctness-oriented culture and the first-answer-to-new-questions deployment both render it a competitor in ours. We re-elaborate this in Section *Research Gap* and Section *Conclusion*.

Reconciling the two sets of findings within one framework.

Our framework accommodates both sets of results without appeal to a separate theory. Grounded in new framework, **we now claim that the effect of an AI answer depends on the AI profile.** Our results confirm this framework. The **AI label alone** effect on supply is positive and significant (0.188, $p<0.05$), the interaction with **AI answer length** is negative (-0.046, $p<0.001$), and the average total effect turns negative only because AI answers in our setting carry substantial content, which is identical to the total deterrence effect in the original manuscript.

The two components behave differently depending on whether the AI is a complement or a substitute. When the AI is a complement, whether because of the community's conversational culture or because it enters an established thread, the label effect is positive, since the AI acts as a catalyst that revives discussion. The length deterrence effect, in contrast, is largely absent. Because the AI and human contributors draw on different strengths, a longer and more complete AI answer does not lead a contributor to conclude that there is nothing left to add. In this case the AIlabel effect dominates and the net effect on contribution is positive, which matches the results of Su et al. (2025). When the AI is a competitor, as in our technical setting where it is the first

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answer to a new question judged on correctness, a longer AI answer signals that the question is already well covered and deters contribution. The content effect dominates and the net effect is negative. Both findings follow from the same two components, and the direction of the net effect depends on whether the AI is read as a complement or a substitute.

Reference

Oliveira, N., Muller, M., Andrade, N., & Reinecke, K. (2018). The exchange in StackExchange: Divergences between Stack Overflow and its culturally diverse participants. *Proceedings of the ACM on Human-Computer Interaction, 2*(CSCW), 1-22.

[1] Examples from actual Quora content. See: https://www.quora.com/What-did-Marilyn-Monroe-carry-in-her-coffin and https://www.quora.com/What-is-the-best-small-business-to-start-in-Gambia

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#R1-6 (source effect)

– I have concerns about the theoretical coherence of Section 3.1, particularly regarding the source effect and the use of dual-process theory.

- *First, the source effect relies on the assumption that users attribute heightened authority and credibility to AI-generated answers, drawing on algorithm appreciation and automation bias. However, this assumption appears to conflict with the institutional context. The platform explicitly restricted external GenAI use due to concerns about inaccuracy. After carefully checking the translated version of user comments under the announcement (https://segmentfault.com/a/1190000043031126), it appears that users expressed substantial skepticism and dissatisfaction rather than unquestioned trust toward AI-generated answers.*

Response

We have looked closely into the tension the reviewer identifies. Before addressing it, we note one change in how the AI effect is organized in the revised manuscript, since this reorganization also underlies our responses to several later comments. In the previous version, we decomposed the total AI effect through a mediation analysis and attributed a residual portion to a source effect. In the current version, grounded in new framework, **we now claim that the effect of an AI answer depends on the AI profile.** Our results confirm this framework. The AI label alone effect on supply is positive and significant (0.188, $p<0.05$), the interaction with AI answer length is negative (-0.046, $p<0.001$), and the average total effect turns negative only because AI answers in our setting carry substantial content, which is identical to the total deterrence effect in the original manuscript. This is a change in how we identify and present the mechanisms, not a reversal of our earlier finding that AI presence deters human contribution.

With this in mind, we agree that the platform context does not support an assumption of unquestioned trust in AI-generated answers. In the earlier version, we treated part of the total deterrence as arising from heightened authority and credibility attached to the AI source. On reexamination, and consistent with the user reactions the reviewer points to, we now read the label differently. In this technical Q&A domain, the bare AI label triggers skepticism rather than

Deleted: We thank the reviewer for this rigorous critique. We have substantially revised our explanation and now organize our differentiation from Su et al. (2023) around three fundamental distinctions:
First, the platforms differ fundamentally in community culture. We agree that Quora contains both objective and subjective contents. We want to emphasize that the two platforms have difference in community culture(Burtch et al. 2024). Quora primarily serves casual, everyday topics. Questions like "What did Marilyn Monroe carry in her coffin?" and "What is the best small business to start in Gambia?" illustrate its conversational orientation.[1] Oliveira et al. (2018) document that Quora exhibits significantly more "conversation ingredients" than Stack Overflow. SegmentFault, like Stack Overflow, operates under an efficiency-oriented philosophy, and see for correct answers for each question. This difference creates distinct contributor incentives when AI providing answers. We re-elaborate this in the paper.
Second, experiment setting differs. We apologize our mis-interpretion of the "three-month period" in the previous version and appreciate the reviewer for pointing this out. In Su et al. (2025) AI answers were provided **to existing questions** through batch processing rather than deployment upon postings instantaneously after their generation. As a result, AI answers may reactivate old questions on Quora and trigger more human participation. In our experiment, AI answers appeared instantaneously after posting. Human answerers respond under this prior condition. This timing difference may trigger different behaviors of human answerers.
Richer Theoretical Finding: The Label Effect Under Different Boundary Conditions
Due to these differences, we find different while complentary results from Su et al. (2025). Specifically, Su et al. (2025) only found the AI label effect in driving human contributions. They did not find the content quality matters.
Going beyond Su et al.'s findings, we not only find AI lable's effect but also find that AI label's effect mostly work with content quality (in a platform seeking for correct answers) in deterring human answerers' participation in an questior ... [1]

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3. D'Angelo, A. (2024). Introducing Poe. Quora Blog. See: https://quorablog.quora.com/Poe-1

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The skepticism and dissatisfaction the reviewer identified in the user comments under the announcement are the sentiment this mechanism relies on, and we now use them as supporting evidence for H1. Because the AI label alone does not by itself deter contribution in this domain, the deterrence we observe operates largely through the answerer's evaluation of the AI length rather than through the label alone. The reviewer's reading of the context is thus incorporated directly into the revised theory.

We hope you find this way of theoretical development under a coherent overarching theoretical framework clearer and satisfactory.

Reference

Tversky, Amos, and Daniel Kahneman. "Availability: A Heuristic for Judging Frequency and Probability." *Cognitive Psychology* 5, no. 2 (1973): 207–32.

#R1-7 (theory – why dual-process)

- *Second, the attempt to integrate the content, source, and position effects using dual-process theories (e.g., central vs. peripheral routes) feels underdeveloped and somewhat stretched. It is unclear why specific effects should operate through particular cognitive routes, and no empirical evidence or established literature is provided to justify these mappings. Dual-process theories typically require clear conditions under which central versus peripheral processing dominates (e.g., motivation, ability, involvement), which are not articulated or tested here. As currently presented, the framework appears post hoc. The authors should either provide stronger theoretical and empirical justification for this integration or simplify the theorization to avoid overextension.*

Response

We have rebuilt the theoretical development section in response to this point. The revised framework no longer maps preexisting effects onto processing routes after the fact. It begins from a single overarching theoretical framework.

The revised framework derives six predictions from three signals across the two decisions. Specifically, we specify which mode of processing governs each decision and state the condition under which it does so. The answerer's encounter unfolds in two stages. The first decision, whether to contribute at all, is binary and fast, and it is governed by System 1 because the only available information consists of surface cues, the AI label, and the AI answer length, which can be processed at a glance and cost almost nothing to process under the law of least effort. The second decision, how much effort to invest, is governed by System 2, and we state the condition for its engagement explicitly. Substantive answer quality is a slow signal that only deliberate reading can disclose, and System 2 engages because the competitive stakes, including a threat to

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the answerer's professional identity, are judged sufficiently important to override the law of least effort.

Each route assignment is tied to a specific, testable hypothesis. For answer supply, the AI label is a fast source cue and predicts a positive level effect in H1, length is a glance-based coverage cue and predicts a negative slope in H2, and quality requires reading and predicts no appreciable slope in H3. For first human answer quality, the label predicts a negative level effect in H4, length predicts no appreciable slope in H5, and AI quality sets a benchmark and predicts a positive slope in H6. The mappings are therefore stated in advance and examined empirically rather than asserted post hoc.

Our results confirm this framework. The AI label alone effect on supply is positive and significant (0.188, $p<0.05$), the interaction with AI answer length is negative (-0.046, $p<0.001$), and the average total effect turns negative only because AI answers in our setting carry substantial content, which is identically the total deterrence effect in the original manuscript.

We hope the revised theoretical development section makes the theoretical grounding clearer.

#R1-8 (quality degradation)

– I find Section 3.2 conceptually problematic.

- *The distinction between the proposed mechanisms, expertise composition shift and reduced effort investment, is unclear. The expertise composition shift assumes that AI answers disproportionately discourage experts while having minimal impact on non-experts, but the paper does not explain why non-experts would not also be discouraged by the same signals of redundancy or elevated standards. The reduced effort investment mechanism further blurs this distinction, as both mechanisms ultimately describe contributor discouragement leading to lower-quality human answers. As currently theorized, these mechanisms are not analytically separable.*

Response

We agree that the two mechanisms, expertise composition shift and reduced effort investment, both ultimately describe lower-quality human answers arising from contributor discouragement. And they are hard to analytically separate given the limitation that we don't have good enough effort-related proxies. Our earlier intention had been to combine them under the single heading of quality degradation and to stop distinguishing whether the change came from a shift in who contributes or from a change in the effort of the same contributors.

In the revised manuscript, this issue no longer arises. We have removed the analyses based on netlikes to make the paper more focused and to ensure the results align more strongly with our updated theoretical framework. Consequently, we have also removed the section that developed those two mechanisms; the current version no longer relies on the distinction the reviewer questioned.

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#R1-9 (elevated expectations)

- *The elevated expectations mechanism again relies on the assumption that users perceive AI-generated answers as superior and norm-setting. This assumption is not well supported in the platform context, where AI use is explicitly restricted due to accuracy concerns and user reactions suggest skepticism toward AI-generated answers. Without clearer justification that AI establishes a credible quality benchmark, the elevated expectations mechanism remains speculative.*

Response

We agree that an elevated expectations mechanism rests on users perceiving AI answers as superior and norm-setting. This key assumption, in our current view, is captured in the H2 of our current theoretical framework. Potential human answerers are deterred primarily by the perceived high coverage of the AI answer, a surface cue that can be inferred from length at a glance. Length is processed through attribute substitution, in which System 1 replaces a difficult judgment with an easier one associated with it (Kahneman 2011; Kahneman & Frederick 2002). The difficult judgment is how much of the question the rival has covered; the easy substitute is physical size. A visibly long AI answer is read as the work of a competitor that has treated the question comprehensively, which is developed and cultivated based on the reviewer's idea about "AI-generated answers as superior".

As stated in our opening remark, to make the paper more focused and to ensure the results align more strongly with our updated theoretical framework, we have respectfully placed this analysis here but removed all netlike-related results, including the elevated expectations mechanism, in the manuscript.

#R1-10 (5 mins)

- *In Section 4.1, the authors assert that AI-generated answers "typically occupy the first position for treated questions" and that viewers "typically see this content before human-generated answers." It is unclear what "typically" means in practice. Does this imply that human-generated answers sometimes occupy the first position despite the presence of AI-generated answers? If so, the authors should report descriptive statistics on answer ordering, such as the proportion of treated questions in which the AI answer appears before any human answer and the fraction of cases in which human answers precede the AI answer.*

Response

Sorry for the confusion. To clarify the underlying mechanism: on this platform, an AI-generated answer is, by design, always positioned before any human-generated answer for a treated question. The AI answer is not interleaved with human answers according to posting time; it is anchored to the top of the answer list.

Due to the time needed for AI answer generation, there is a small delay. In practice, this window is negligible relative to human response latency, since the first human answer is posted 22.067 hours on average. It will be faster than most humans.

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#R1-11

– I noticed inconsistencies in the reporting of key covariates across the main tables and the appendix. For example, Table 2 reports the average question length for the treatment group as 84.233 Chinese characters, whereas Table A1 reports a mean "log number of Chinese characters" of 0.004; similarly, Table 2 reports an average questioner reputation of 334.484, while Table A1 reports a mean "log reputation score" of 4.026. These values are not consistent under standard log transformations.

Response

We sincerely apologize for this confusion. Upon careful review, we identified a formatting error in the Appendix where values were misaligned across columns. We have thoroughly corrected all such inconsistencies. Thank you for your careful reading that helped us identify this issue.

#R1-12 (only length proxies)

– Section 6.3.1.1 uses "first answer length" as a key variable in testing the content effect, but the definition is unclear. In treated questions, does this refer to the length of the AI-generated answer (which is typically the first answer), or to the length of the first human answer? These two interpretations have very different implications for the mechanism test and for causal interpretation, especially given that human answer length would be post-treatment. The authors should clearly define this variable and justify its use in the analysis.

Response

We clarify the definition first. First answer length refers **to the length of the first answer posted under a question, and this variable is for both the treatment group and control group.** Because the AI answer is posted first in treated questions, this first answer is the AI answer in the treatment group and the first human answer in the control group. We use it to control for the quality of the first-position content that a later answerer sees before deciding whether and how much to contribute. We note that this variable was required by the analysis strategy used in the previous version, in which controlling for first-answer content was necessary to isolate the effect of interest, and this is the reason the same variable appeared on both treated and control questions.

In the revised version, we have changed the analysis so that the content variable is defined directly as the quality of the AI content, taking the value zero when no AI content is present, which is the case for all control questions. This specification does not mix AI and human content within a single variable, and it lets us separate the AI label effect from the AI content effect, since the label distinguishes treatment from control while the AI content quality varies within treated questions. We have updated Section Results to state this definition and to explain the change.

Beyond the definition, we see the comment as raising a substantive measurement question, namely whether a single length variable can adequately capture content quality. We agree that length alone is a limited proxy, and this concern is the reason we had earlier introduced an LLM-based quality score. In this revision we took two further steps to address it directly.

First, we clarify that answer length measures coverage rather than overall quality. In our original framing, we imprecisely characterized length as a quality proxy. We now position it

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specifically as a measure of answer coverage, which captures the breadth and thoroughness of comprehensiveness. This is conceptually distinct from the dimensions captured by our original LLM-based measures. As detailed in Appendix B, the LLM assessment focuses on objective quality dimensions such as accuracy, clarity, and relevance, which evaluate how well an answer addresses what it chooses to cover rather than how much ground it covers. This distinction explains the divergent mediation patterns. Potential human answerers are deterred primarily by the perceived high coverage of the AI answer, a surface cue that can be inferred from length at a glance, rather than by fine-grained quality, which becomes available only through effortful evaluation.

Second, we highlight the 3 validation tests we conduct for the LLM-based quality measure. Failed AI answers (for example, "I'm sorry, I can't provide a detailed answer") received significantly lower quality ratings than substantive answers ($p < 0.001$). Human answers accepted by questioners received significantly higher ratings than non-accepted answers ($p < 0.05$). Answers with above-median or above-mean net likes received significantly higher ratings than those below the median ($p < 0.001$). These results support the validity of the measures.

Third, we examined what answer length actually captures. Length can reflect perceived quality, or it can reflect perceived coverage, meaning the impression that an answer spans many aspects of the problem. To separate these two readings, we added a second content variable, the number of bullet points in the answer. GenAI answers in this setting are typically well formatted and well structured, and an answer that enumerates several points tends to convey broad coverage, which motivates a human perception of comprehensiveness. If length were operating mainly through perceived coverage, its effect should track this coverage variable rather than a direct quality signal. Our results indicate that the role of length does operate largely through perceived coverage and not through perceived quality. We report this analysis in mechanism part.

#R1-13 (position effect)

> – *I find the logic surrounding "first-position seekers" in Section 6.3.1.1 unconvincing. The authors argue that when an AI answer occupies the first position, first-position seekers are less likely to provide the first human answer. However, even in the presence of an AI answer, users can still be the first human responder, so it is unclear why this opportunity is meaningfully reduced. This interpretation implicitly assumes that first-position seekers care only about being absolutely first rather than first among human contributors, which is neither theoretically justified nor empirically demonstrated.*
>
> *Moreover, the finding that the presence of a first-position seeker strongly stimulates subsequent participation is counterintuitive. Early answers often satisfy demand and crowd out later contributions rather than encouraging them. The paper does not explain why first-position seekers would systematically generate answers that invite further engagement. Are answers from these first-position seekers of low quality?*

Response

We address the two parts of this comment in turn before explaining the change we have made.

On the first part, the reviewer correctly identifies our implicit assumption. We did assume that

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first-position seekers care about being absolutely first rather than first among human contributors. This assumption follows from our competitive framing. Because the human answerer and the AI compete for the same scarce resource, the limited attention and recognition of viewers, the AI answer is a competitor for the first position and not merely background content. From this standpoint, an existing AI answer reduces the value of the first human answer for a first-position seeker.

On the second part, regarding why the presence of a first-position seeker would stimulate rather than crowd out later participation, one reading is that first-position seekers optimize for speed rather than comprehensiveness, so their early answers tend to be narrow in coverage and leave room for others to add what is missing, while also signaling that the question is an active contest worth entering. Whether these early answers are of low quality is one possibility the reviewer raises. We regard it as plausible but not the only account, because an early answer may be narrow in coverage rather than low in quality, and without perception data we cannot cleanly separate these two readings.

This limitation is the reason for our change. We acknowledge that the position effect is difficult to test adequately without data on how answerers perceive position. As stated in R1-4, to make the paper more focused and to ensure the results align more strongly with our updated theoretical framework, the position effect has been removed from the manuscript in this revision.

#R1-14

> *– In Section 6.3.1.3, the authors state that they control for the position effect when estimating the source effect, but it is unclear what specific variables constitute the "position effect controls" and how they capture answer ordering or visibility.*
>
> *More importantly, the claim that a remaining significant coefficient on the AI indicator constitutes evidence of a source effect is too strong. A residual AI effect could reflect other unmodeled mechanisms correlated with AI presence, such as user discouragement or frustration induced by the platform's restrictive policy toward external GenAI and negative user reactions to AI deployment. Without ruling out such alternative explanations or independently manipulating source cues, the residual treatment effect cannot be uniquely attributed to a source effect.*

Response

We thank the reviewer for both parts of this comment, and we address them in turn.

On the "position effect controls." In the earlier version, the position effect was not captured by a direct measure of answer ordering or visibility. Instead, we tested it indirectly through a proxy based on user behavior: for each treated question, we examined whether the first human answer came from a "first position seeker," an answerer identified from pre-experiment revealed preferences as someone who tends to compete for the first-answer position. The logic was that if AI presence deters these users precisely from providing the first human answer, this would provide indirect evidence of a position effect. We adopted this behavioral proxy because a more direct test was not feasible in our setting. Ideally, one would exploit variation in the AI answer's position, but such variation does not exist here: the AI answer is essentially always placed first, and even for the rare "typical" cases we do not observe a precise timestamp for when the AI answer appears, so we cannot construct an answer-position variable. On reflection, we agree with

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the reviewer that this behavioral proxy was too indirect to convincingly capture answer ordering or visibility. This limitation, together with the concern the reviewer raises separately in R1-13 that the position effect is difficult to test adequately in our setting, led us to remove the position effect from the framework rather than retain a mechanism we could not test rigorously. As a result, the "position effect controls" no longer appear in the revised paper, and the framework is now organized around effects we can examine directly.

On the residual logic and the alternative explanation. We agree that treating a remaining significant coefficient on the AI indicator as evidence of a source effect claims more than the design can support, since a residual effect can reflect mechanisms correlated with AI presence that the model does not separately capture.

The revised framework no longer rests on this residual logic. As described in our response to R1-6, the total AI effect is now organized into an AI label effect and an AI content effect, each derived from the fast-and-slow account and examined directly rather than recovered as what remains after other effects are removed.

The alternative explanation the reviewer raises, discouragement or skepticism induced by the platform's restrictive stance toward external GenAI and the accompanying user reactions, is now part of the mechanism we model rather than a confound we need to rule out. In the theoretical development part (H1), the AI label activates this domain-specific skepticism through the availability heuristic (Tversky & Kahneman, 1973), and this skepticism is what leads the answerer to appraise the AI as a weak competitor. What the earlier version would have made into an unexplained residual is therefore given an explicit theoretical role in the current version.

Reference

Tversky, Amos, and Daniel Kahneman. "Availability: A Heuristic for Judging Frequency and Probability." *Cognitive Psychology* 5, no. 2 (1973): 207–32.

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#R1-15

> *– I recommend reporting the results of the causal mediation analysis in a table and provide enough details about the implementation.*

Response

The required detailed causal mediation results in Table R1-15. We understand that the "formal causal mediation analysis" refers to the approach developed by Imai, Keele, and Tingley (2010). We did conduct this analysis in our previous revision, but we acknowledge that the presentation was not sufficiently detailed. In this response letter, we report the full causal mediation results in a dedicated table, with the average causal mediation effect (ACME), average direct effect (ADE), total effect, and the proportion mediated. The results confirm that the content channel accounts for approximately 26% of the total treatment effect.

Table R1-15. Causal Mediation Analysis Results

	Content Effect	Position Effect
Effect Decomposition		

ACME	**-0.041***** **[-0.063, -0.018]**	**-0.031***** **[-0.037, -0.024]**
Direct Effect (ADE)	-0.115*** [-0.144, -0.083]	-0.125*** [-0.144, -0.104]
Total Effect	-0.156*** [-0.175, -0.136]	-0.156*** [-0.176, -0.136]
Proportion Mediated		
% of Total Effect Mediated	**26.50%** **[23.6%, 30.4%]**	**19.70%** **[17.5%, 22.6%]**

But in the revised manuscript we no longer conduct a causal mediation analysis, and we no longer state hypotheses about mediation. The current approach is organized around theory. We derive the hypotheses H1 through H6 from the fast-and-slow account and then test each of them, so that the empirical work supports the theoretical development directly. For this reason, a causal mediation results table is no longer applicable.

Deleted: Thank you for this constructive suggestion. We have now included a dedicated table (Table [X]) presenting the results of our causal mediation analysis, along with expanded methodological details in Section [X] that specify the estimation procedures, identifying assumptions, and bootstrap inference approach employed.

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#R1-16 (comment sentiment)

– In Section 6.3.2.1, the authors use comment volume to rule out attention diversion. While this is informative, under the elevated expectations mechanism, comments may become more critical or negative rather than fewer. Comment sentiment, not just volume, would therefore be more informative.

Response

We thank the reviewer for this insightful suggestion. The logic that elevated expectations may manifest through more critical or negative comments, rather than fewer comments, is compelling. We conducted the suggested analysis, though two data limitations constrain its statistical power. First, platform observations reveal minimal sentiment variation in comments on this technical Q&A community; users typically focus on functional clarifications rather than evaluative judgments. Second, a substantial proportion of answers receive zero comments, and this analysis conditions on the subset with at least one comment, which further reduces sample size.

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Despite these constraints, we constructed two sentiment measures using established Chinese NLP tools. For SnowNLP, which returns a continuous sentiment score between 0 and 1, we classify a comment as negative when its score falls below 0.5. For CnSenti, which separately counts positive and negative sentiment words within each comment, we classify a comment as negative when its negative word count exceeds its positive word count. As shown in the Table R1-16 below, the coefficients on AI are positive under both measures, suggesting a directional pattern consistent with the elevated expectations mechanism, though neither reaches statistical significance.

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Table R1-16. Effect of AI Answers on the Number of Negative Comments Received by Human Answers

	(1)	(1)
DV:	*negCommentNum_cnsenti*$_i$	*negCommentNum_snowNLP*$_i$
AI_i	0.061	0.001
	(0.037)	(0.084)
other controls	√	√
answer timing FE	√	√
N	1,451	1,451

We interpret these positive coefficients as directionally supportive evidence that viewers may express more critical sentiment toward human answers in the presence of AI, consistent with elevated evaluative standards.

As stated in our opening remark, to make the paper more focused and to ensure the results align more strongly with our updated theoretical framework, we have respectfully placed this analysis here but removed all netlike-related results in the manuscript.

#R1-17

– A broader concern is that several analyses throughout the paper restrict attention to subsamples of questions with at least one human answer. Because the main treatment effect is a reduction in human answers, conditioning on having an answer constitutes post-treatment selection. This restriction disproportionately excludes treated questions and selects on the outcome variable, potentially biasing estimates and complicating interpretation of mechanism tests. The authors should clearly justify these sample restrictions, discuss the resulting selection, and assess whether the conclusions drawn from these subsamples generalize to the full set of treated questions.

Response

Yes, in the previous version, the mechanism tests restricted attention to questions with at least one answer.

This restriction was required by the analysis strategy at the time, since controlling for first-answer content is only possible on questions that already have a human answer. We also re-reported the main results on this restricted subsample (in the original manuscript), and the deterrence effect there is larger, which indicates that the deterrence is not an artifact of the sample restriction and that the main results continue to hold.

More to the point, the current version no longer uses this setup in any of its analyses, and the main results do not depend on it. We hope the reviewer will refer to the results section of the revised manuscript, where this concern should be substantially reduced.

#R1-18

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Commented [DC1]: Dear Ben,

This may concern our human 1 quality DV analysis.

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– Page 19 of the revised manuscript, there is a typo: “Summary StatisticsTable 2 “.

Response

Thank you for catching this typographical error. We have corrected it and conducted a comprehensive proofreading of the entire manuscript to ensure no similar issues remain.

#R1-19

I hope the authors find these comments helpful. Best of luck!

Response

We are deeply grateful for your thorough and constructive review. Your comments have been invaluable in strengthening both the rigor and clarity of our manuscript. We sincerely appreciate the time and effort you have devoted to improving this work.

Reviewer 2 Comments

#R2-0

Thank the editors for inviting me to review the revised version of this paper. Thank the authors for addressing my earlier comments.

I am the R2 in the last round of review. My process for this round review is as follows.

First, I scan the authors' responses.

Second, I re-read the revised paper carefully.

Third, I carefully re-read each of my earlier comments and each response from the authors, and consider whether my comments are addressed. If they are addressed, I label them as addressed. If not, I will provide further elaboration/comments.

To organize the work, I followed the same comment label from the authors.

Since this is a revised paper, and I read the paper to understand how my earlier comments were addressed, I, therefore, have some new comments. At the same time, this is a revision. Hence, I keep new comments to a minimum.

I hope my review process makes sense to the authors and editors. I hope the authors the best of luck in improving their paper.

Response

We sincerely thank Reviewer 2 for the systematic and constructive comments. We have carefully revised the paper according to the reviewer's comments. We hope the reviewer finds the paper satisfactory this time.

In this round of revision, we benefited greatly from extensive communication with the **Senior Editor, including both email correspondence and offline face-to-face discussion**. Following SE's **detailed and actionable guidance**, we have made the following major changes. **First**, we substantially refined the theoretical framework along the competition direction suggested by the Senior Editor, establishing the theoretical hypotheses under a coherent and overarching logic. **Second**, we realigned our empirical results with this framework, which involved both adding results that speak more directly to the theory and removing those that were less central to it. **Third**, we sharpened the positioning of our contribution by more explicitly articulating how our study differs from prior work. We elaborate on each of these changes in our point-by-point responses below.

Before responding to individual comments, we describe two major theoretical and empirical changes that run across our responses. Later comments refer back to this note rather than repeating it.

First, a reorganization of how the theory is presented. We have rebuilt the theoretical development based on the email communication and offline face-to-face instruction with SE. It begins from: **1) a competitive encounter perspective** motivated by competition dynamics theory (Chen et al., 2007; Chen, 2009), which explains how the member appraises the AI rival; and **2) a single overarching account Kahneman's fast-and-slow theory** (Evans & Stanovich,

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2013; Kahneman, 2003, 2011), which explains which signal can affect each decision and when that signal becomes available..

Grounded in new framework, we now claim that the effect of an AI answer depends on the **AI profile**. The AI label alone signals that a competitor has entered the question. In a community that already views AI-generated content with suspicion, users treat this competitor as a weak one, and the presence of a weak competitor motivates human contribution (H1). The AI length works in the opposite direction. When the AI answer is long and appears comprehensive, potential answerers judge that little of value remains to add, and their contribution falls (H2). The positive prediction is thus derived from theory before analysis and is stated in the hypotheses themselves, not inferred after the fact.

Our results confirm this framework. The AI label alone effect on supply is positive and significant (0.188, $p<0.05$), the interaction with AI answer length is negative (-0.046, $p<0.001$), and the average total effect turns negative only because AI answers in our setting carry substantial content, which is identically the total deterrence effect in the original manuscript.

Second, a deliberate narrowing of scope to keep the story focused. The removal is a choice about narrative focus and length, not a retraction of the earlier logic. We removed three elements from the main text. (1) All results that use net likes as the dependent variable, together with every recognition mechanism built on them, are removed. (2) The position effect is removed from the main text. (3) In the response letter, we still engage fully with the reviewer's earlier concerns about these removed elements, in order to show that the original reasoning was sound.

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#R2-1

Theoretical Foundation

R2-1. In the earlier round, I asked why the authors focused on content effect, source effect, and position effect instead of other theoretical mechanisms such as the learning effect and the crowded-out effect.

The authors responded by showing that the literature reveals that the three effects are used to explain how answers (earlier answers) affect subsequent user participants, attention, and perceived credibility. And the authors integrate their knowledge of the research context with those established theories/literature findings.

It may not be a good way to justify the selection of the theoretical mechanisms. Other mechanisms, such as the learning, crowd-out, anchoring effect, and cognitive offloading (can be more), are also examined in the literature and can be used to explain this research context (i.e., AI answers' effect on human answers).

Probably, the authors can find an overarching theoretical framework to support their exploration of what the AI says (content effect), who says it (whether AI or human; source effect), and when/where the AI says it (position effect).

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Response

We thank the reviewer for pushing us on the choice of mechanisms. We agree that learning, crowding out, anchoring, and cognitive offloading are all relevant to this setting. In this revision we do not treat them as competing alternatives that we exclude. Instead, we adopt an overarching framework, Kahneman's dual-process theory, that organizes these forces and specifies which mode of processing governs each of the answerer's decisions (Evans & Stanovich, 2013; Kahneman, 2003, 2011). Please see the revised theoretical development section.

The reviewer's suggested mechanisms map directly onto our framework rather than sitting outside it. **Learning** corresponds to the deliberate, System 2 engagement in H6, where an answerer who reads a high-quality AI answer raises the quality of their own contribution. **Cognitive offloading and the crowding out** of effort operate through the law of least effort that governs our entire fast stage, and they enter the theoretical framework through H2, where a comprehensive AI answer changes the perceived cost and benefit of contributing. **Anchoring** is now the explicit core of H4: the weak competitor judgment formed from the bare AI label acts as a kind of anchor on subsequent effort. Having judged the rival to be weak, the member expects that even a fairly modest answer will be enough to come out ahead. The law of least effort holds that people supply effort only up to the level a goal requires, and no further (Kahneman 2011)

Adopting a single overarching framework, rather than a list of separate mechanisms, lets us explain the two decisions an answerer actually faces, namely whether to contribute and how much effort to invest, within one coherent account. Following this reorganization, the "when and where the AI appears" dimension is no longer part of the framework. As noted in our opening remark, the position effect has been removed from the main text to keep the theoretical story focused on the fast and slow processing of AI cues.

#R2-2

R2-2. In the last round, I asked whether it is theoretically possible that AI answers increase human participation and provide a "feeling that the authors get some empirical findings first and then try to propose some mechanisms to explain the findings."

The authors responded by clarifying that their "theoretical framework was developed before conducting analyses."

Based on the authors' response, "We appreciate your suggestion that AI answers might increase human participation by providing a "starting point." While theoretically possible, this mechanism is not supported by our empirical findings, which provide evidence against such a "starting point" mechanism." the authors admitted the theoretical possibility that AI answers increase human participation. And they choose not to propose it because of their empirical findings.

To summarize, the authors should discuss the theoretical predictions of the positive effect of AI answers on human participation in the front unless they argue that it is not theoretically possible. Using empirical results to guide theoretical propositions is HARKing (Hypothesizing After Results are Known; Kerr, N. L., 1998), as the authors highlighted.

Response

We fully accept the reviewer's point that a theoretically possible positive effect must be theorized in the front matter. Our revised framework now does exactly this.

H1 formally predicts a positive effect of the AI presence on human contribution. When an

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answerer sees the AI label alone, the availability heuristic retrieves a domain-specific impression that AI is unreliable on genuine technical problems, and this produces a weak competitor judgment (Tversky & Kahneman, 1973). Appraising the AI as a weak competitor pushes toward entry. The AI answer sits under the same question, faces the same readers, and competes for the same attention, precisely the condition under which an actor is most likely to notice a rival and respond (Chen et al. 2007; Chen 2009). The mere presence of a competitor in the same thread is an invitation to respond, and a weak competitor makes responding attractive: the risk of losing is low, and a competent developer can supply the correct answer the AI failed to supply. H1 therefore predicts that the AI label alone, increases contribution.

The observed net reduction in human answers is the sum of two opposing forces that are both theorized in advance: the positive AI label effect in H1 and the negative effect of AI answer length in H2, with the length-driven force dominating in our setting. The positive prediction is thus derived from theory before analysis and is stated in the hypotheses themselves, not inferred after the fact. This structure directly resolves the HARKing concern.

Deleted: Thanks for the suggestion. We appreciate the reviewer's suggestion of human participation. But it is within our theretical framework. We acknowledge that there exist multiple parallel theretical framework that aim to explain one phenomenon. But we choose the one that is more plausible than the one proposed by the reviewer.
We apologize for not being cystyle clear in the last review that we do not think AI answer can increase human participation. In the revision, we highlight XXX theory as the overarching theory to direct our modeling, of which human participation is not a major factor.

#R2-3 (content effect)

> *R2-3. I provided a counterargument for the content effect by arguing that the imperfections of AI can perform worse than humans. The authors argued that "If AI-generated answers are sufficiently comprehensive and well-articulated to satisfy the immediate information needs of potential answerers and questioners, they reduce the perceived gap between existing information and what potential contributors might add, which in turn decreases contribution motivation. Conversely, if AI answers are of insufficient quality, they would not generate this crowding-out effect. Our empirical findings indicate which scenario prevails in our setting." Additionally, they provide empirical evidence to show that AI answers in their context are sufficiently comprehensive.*
>
> *I feel my concerns were not addressed, and the counterarguments hold.*
>
> *I agree with the authors that if AI performs better than humans, some humans who can not make new contributions may decrease their motivation to contribute. However, the authors didn't argue theoretically that AI performs better than humans in this context; they just said that "If AI-generated answers are sufficiently comprehensive…"*
>
> *Although they provided some empirical evidence (in the empirical sections) to support that AI performs better than humans, the theoretical arguments about the superiority of AI answers are missing.*

Response

We thank the reviewer, and we agree that our earlier phrasing relied on an empirical "if the AI answer is sufficiently comprehensive" clause without a theoretical argument for why the AI answer would be strong enough to matter. We now supply that argument in the revised Section 3, and we deliberately avoid the stronger and harder-to-defend claim that AI outperforms the best human answerer.

Why the AI answer is strong enough to matter. We do not claim that AI dominates all humans. The platform's human contributors form a certain distribution of ability, and while AI's quality may not exceed that of the best human, it certainly exceeds that of the weaker contributors;

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otherwise the platform would not have deployed the system at all. This is sufficient for a "crowding-out" effect: the AI answer will deter the segment of potential rookie contributors who think AI is not weak in their mind, because they think the answers the AI has already supplied what they could add. This is the basic argument for our content effect.

Why perceived AI quality deters contribution: perceived coverage as a fast signal. Beyond this logic, we also theorize the perceptual channel through which AI content operates, since what deters a potential contributor is the *perceived* rather than the objectively verified quality of the AI answer. Consistent with the fast-and-slow account, potential contributors do not systematically evaluate the AI answer before deciding whether to contribute; instead they rely on readily available surface cues. AI answers are characteristically long and heavily structured into bullet points, and this length and structure serve as a heuristic (fast) signal of high coverage. An answer that *looks* thorough raises the potential contributor's perception that the information need has already been met, which lowers the perceived gap that a new human answer could fill and thereby deters contribution. This is why the content effect operates through perceived comprehension, and it is reflected in our empirical test of AI answer length and AI answer quality in the results section.

For our current treatment of the content effect, we respectfully direct the reviewer to the revised Section 3, Theoretical Development, in particular H2 and H4. We hope this fuller theorization addresses the concern.

#R2-4 (source effect)

> *R2-4. I countered the source effect by addressing potential concerns about attributing greater authority or credibility to the AI source, since the authors used the literature on predictive AI to argue for generative AI and avoid discussing algorithm aversion.*
>
> *In response, the authors argued that algorithm appreciation and aversion are not mutually exclusive, and they posit that "within online technical Q&A communities, the AI source label carries specific credibility signals that influence user behavior." In addition, they argued that there are common contexts between the forecasting contexts (used in Logg et al. 2019) and this research context.*
>
> *My concerns still hold. The authors didn't address whether the findings in the literature about the authority/credibility in predictive AI in context, such as forecasting contexts (used in Logg et al. 2019), work, and are suitable to argue for the authority and credibility of the generative AI. Even though both predictive and generative AI are part of AI, they work differently. Generative AI used in this research can fabricate and hallucinate. Whether users hold the same level of authority and credibility perception on them as they do on predictive AI is unknown.*

Response

We appreciate this comment and have made three changes in response.

First, we have removed the analogy to predictive AI. We agree that findings on authority and credibility in forecasting contexts, such as those in Logg et al. (2019), cannot be transported directly to generative AI, which can fabricate and hallucinate and which users may appraise very differently.

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Deleted: Thanks for raising this comment. While the AI system of the platform can be considered as have a certain quality level, the human contributors of the platform forms a certain distribution. AI's quality may not be better than the best human, it is certainly better than the worst human. Otherwise, the platform won't need of setup the system. Thus, given a certain AI level, its content will for sure crowd out some users. This forms the basic argument for our content effect

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Deleted: We appreciate the concern of the reviewer. In the revision, we removed the analogy to predictive AI. Instead we directly attribute the effect to AI fantasy caused by GenAI in these years. [Find some literature]

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Second, on reflection we accept the reviewer's underlying logic about the source. Our earlier notion of an "illusion of authority" was our attempt to name where the overall deterrence came from. On closer analysis, an illusion of high-quality perception could only arise after an answerer has glanced at the AI content and found it comprehensive. Any such effect therefore belongs to the content level, not to the bare AI label alone that an answerer sees before reading content. We no longer attribute a source-based authority effect to the bare AI label.

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Third, the effect of the bare AI label is now explained without any appeal to predictive AI. When an answerer sees only the label, the availability heuristic retrieves the prevailing domain-specific impression in developer communities, namely that AI-generated answers remain unreliable and error-prone on genuine technical problems (Tversky & Kahneman, 1973). This produces a weak competitor judgment and motivates contribution, as theorized in H1. This account is fully consistent with the fact that generative AI can fabricate and hallucinate, since the answerer's doubt is precisely about the AI's reliability on the technical problem at hand.

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We sincerely invite the reviewer to Section 3.1 where the AI label alone effect is theoretically developed. We hope you find this way of theoretical development under a coherent overarching theoretical framework clearer and satisfactory.

#R2-5 (position effect)

> *R2-5. I provided some counterarguments for the position effect. Specifically, I argued the position effect is negligible because questioners and answerers know the first answer in the treatment group is AI. Therefore, the answerers will work hard to secure the first human position (i.e., the second position in the treatment group). Since both groups of users are randomly assigned, we can assume them to have similar sensitivity to become the first human answerers. Therefore, the position effect is similar across both groups and negligible.*
>
> *In response, the authors used three points. Position effect is grounded in the literature. Theorizing the position effect does not include the AI source. Empirical evidence shows that AI answers reduce human participation.*
>
> *My concerns still hold. First, I agree that the position effect exists. Second, I think arguing the position effect has to consider the AI source because the first position in the treatment group is the AI answer, and AI answers work very differently for both the behavior of questioners and answerers from human answers on the platform. According to the authors, the questioners and answerers cannot engage with the AI's answers (e.g., "accept", "like", "comment"). It makes the first position as the AI or the human answers work differently. Arguing the position effect without accounting for the different mechanisms by which AI and humans work does not make sense.*
>
> *Third, empirically, AI answers appearing first reduce human participation. This empirical evidence does not mean that answerers tend to contribute less because they can not get the first-position answers. Those mechanisms that the author proposed can explain the reduction effect. In addition, it may be that the AI answers squeeze out simple answers, making those who tend to provide simple answers not contribute.*

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Response

We thank the reviewer for these detailed counterarguments. We read the comment as raising three distinct concerns, and we address them in turn.

Concern 1: Answerers will compete for the first human position, so the position effect is similar across groups. We did assume that first-position seekers care about being absolutely first rather than first among human contributors. This assumption follows from our competitive framing. Because the human answerer and the AI compete for the same scarce resource, the limited attention and recognition of viewers, the AI answer is a rival for the first position and not merely background content. From this standpoint, an existing AI answer reduces the value of the first human answer for a first-position seeker.

Concern 2: Theorizing the position effect must account for the AI source, because AI and human answers function differently. We find this point reasonable to a meaningful extent - the first position in the treatment group is occupied by an AI answer. This consideration, together with 1) the purpose of making the story focused and the limitation of pages; 2) the difficulty of adequately testing the AI position effect in our setting (a limitation we also discuss in relation to R1-13/R1-14), led us to remove the position effect from the framework. Importantly, the concern the reviewer raises is not lost in the revised theory. What was previously the source effect is now developed as the **AI label effect**, and its development already presumes that the AI answer is the first mover occupying the dominant first position and is therefore processed as a fast, heuristic signal. In other words, the interaction between position and source that the reviewer emphasizes is now internalized in the AI label effect rather than handled as a separate position mechanism.

Concern 3: The empirical pattern does not uniquely establish a position mechanism. We agree. As stated in our opening remark, we have removed the position effect from the main text in this revision, in order to keep the theoretical story focused on the fast and slow processing of AI cues. Because the position effect is no longer part of the paper, we do not further develop the argument about the position effect.

We are grateful that the reviewer's point helped clarify the boundary of what our focused framework now claims. As stated in our opening remark, to make the paper more focused and to ensure the results align more strongly with our updated theoretical framework, the position effect has been removed from the manuscript in this revision.

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#R2-6 (expertise composition shift)

R2-6. I provided a different view of the expertise composition shift mechanism. I argued that the experts should remain in the treatment group because they have domain knowledge, enabling them to leverage AI-generated answers to improve their own answers. Additionally, the answerers will tend not to contribute instead of contributing a lower-quality answer when they see an AI answer. Thus, the human answers in the treatment group should not show a decreased viewer recognition.

In response, the authors argued: 1. "The key issue is that expert participation depends on perceived marginal contribution, not merely on their capacity to differentiate." 2. "comprehensive AI answers dramatically reduce the perceived unique value experts can add." 3. "Thus, rational experts will reduce their effort investment in crafting high-quality answers to these questions, leading to the expertise composition shift that we document."

My concerns still hold. First, the authors admitted that expert participation depends on their capacity to differentiate (see the first point in the previous paragraph). Therefore, solely

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arguing the marginal contribution without arguing the expert's capacity in leveraging AI answers to improve their likelihood to stay and improve their missed answers misses the important pieces. Overall, AI answers can provide a good starting point for experts and save the cognitive costs to improve their answers.

Second, the authors assumed that the AI provides comprehensive answers in their second point. Whether AI can provide comprehensive answers depends on many factors, including whether the platform fine-tuned their internal generative AI, what is their in-built prompt, and whether they tested it using experts as the feedback providers.

Third, my original comment is "In my view, humans are rational, and if they find they can not get recognition for their contribution, they will not contribute (as indicated by the authors, "Anderson et al...") instead of "becoming disinclined to invest substantial effort in crafting high-quality answers." I meant that if humans feel they can not get recognition, they will not provide a human answer, which makes the remaining human answers have a higher likelihood of getting recognition. It is because only those humans who believe they can get recognition after seeing an AI answer. Therefore, I suggest arguing the positive side of the AI answer on the viewer recognition of human answers.

Response

We thank the reviewer for this careful alternative. We first address the substance and then note the change in scope.

On the substance, our account rests on marginal contribution rather than on absolute answer quality. Viewer recognition rewards what an answer adds beyond what is already available, not the answer's quality in isolation. If recognition tracked absolute quality alone, a competent answer that merely restated an existing high-quality answer would also earn recognition, which is not what communities reward. We agree with the reviewer that expert participation depends on the capacity to differentiate. Our point is that this capacity is bounded by the AI answer already present. When a highly comprehensive AI answer already occupies the field, even a domain expert finds little room to add differentiated marginal content, so the marginal contribution of an equivalent human answer falls, and recognition for it falls with it. This is consistent with the reviewer's observation that only answerers who believe they can still earn recognition will contribute. Precisely because the comprehensive AI answer compresses the space for differentiation, the marginal value that experts can add is reduced, and this is what drives the shift we described.

On scope, and as stated in our opening remark, to make the paper more focused and to ensure the results align more strongly with our updated theoretical framework, this revision removes all results that use net likes as the dependent variable, together with the recognition mechanisms built on them. The expertise composition shift and the elevated expectations mechanism are therefore no longer part of the main text. We retained the exchange above so that the reviewer can see the reasoning was sound, and we removed the results only to keep the paper focused.

#R2-7

Empirical Analysis

R2-7. I raised a concern about the data's validity because the average number of questions

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[Marginal utitly theory?]

per day between 2021 and April 2024 is 65, while the experimental period (Sept. 2023 – Feb. 2024) only has 24 questions per day.

In response, the authors claimed that "This difference does not indicate a data validity issue. Rather, it reflects the natural evolution and variation in platform activity over time." and the "observed activity level during our study period reflects the platform's actual usage patterns at that time."

I am still concerned because the authors did not explain in detail the steps they took to verify their data. Instead of checking the data they have, they should consult with the platform owners. If it is possible, the authors should consult with the platform owners about the potential reasons and ask the platform to share the evolution of the number of questions on the platform across different months. This reason behind the big difference (65 questions versus 24 questions per day) can potentially confound the treatment effects. I bet the platform owners know/want to know the reasons behind as well.

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Response

We thank the reviewer and have taken concrete steps to be more transparent. SegmentFault posts all questions publicly, so we were able to reconstruct the full time series of question volume directly from the platform rather than relying only on aggregate averages. Figure R2-7 plots the number of questions and answers per day across the platform's entire history. The plot reveals an initial growth phase followed by a gradual, monotonic decline in activity. This pattern is characteristic of a maturing community: as the repository of existing knowledge grows, users are increasingly likely to find answers to their queries in previous posts, naturally reducing the volume of new questions. This trend reflects the community's organic evolution rather than any anomaly during our study window. The figure of 65 questions per day represents the average over the full multi-year span, whereas the figure of 24 reflects the lower activity level during the later experimental window of September 2023 to February 2024. These two numbers are consistent with a single, long-term declining trend rather than a data quality problem.

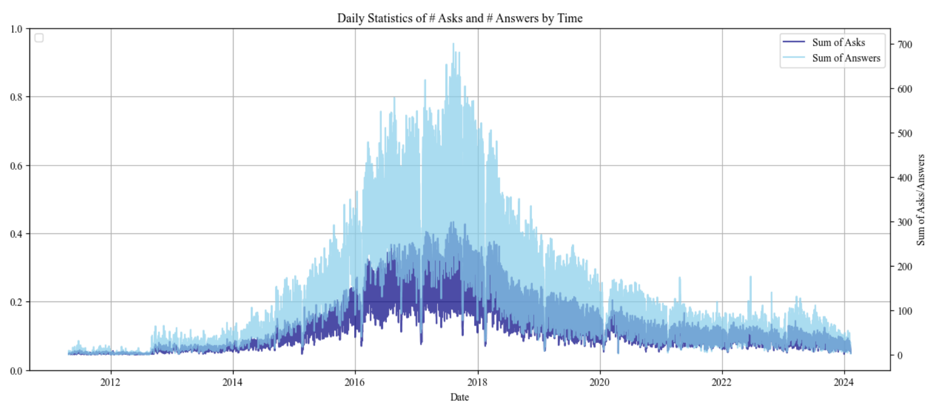

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Figure R2-7. The Number of Questions and Answers per Day Across the Platform's Entire History

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This declining trend does not confound our estimated treatment effect. Treatment was randomized at the question level within the experimental window, so any secular change in platform-wide activity applies equally to treatment and control questions and differences it out. In addition, all specifications control for question timing FE or a cubic polynomial of question age, which absorbs smooth time trends within the window.

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#*R2-8

R2-8. In the last round, I questioned the inconsistency of the sample size. I want to elaborate on my concern in more detail. In the last round of submission, the following text captured my attention.

"From the 3,613 questions in the treatment group, we identify 993 questions with an LLM-enabled content quality rating in the top 25% quantile (scores ranging from 8 to 10), indicating the superior quality of the first answer. Conversely, another 993 questions display quality ratings in the bottom 25% quantile (ratings ranging from 0 to 6), denoting the low quality of the first answer."

There are two inconsistencies. First, there are only 1,766 questions in the treatment group instead of 3,613.

Second, there are 993 questions in the treatment group with the top 25% quantile of their first answer, and another 993 questions in the treatment group with the bottom 25% quantile of their first answer. Only considering those top 25% and bottom 25%, the total number of questions in the treatment group is 1,986, larger than 1,766.

In response, the authors claimed that the total number of questions in the treatment group is 1,766, and the 993 is the matched pairs (993 questions with AI answers matched to 993 questions without AI answers).

I also read carefully about the revised section (section 6.3.2.2), highlighted by the authors. I found my concerns were not addressed. The authors just rephrased the above-quoted paragraph by removing the numbers and avoiding disclosing the number of questions in the top 25% and bottom 25% subsamples in the analysis. It is concerning.

For my first concern, it can be a mistake on the authors' side in treating the 3,613 questions as treated questions. But the authors need to respond to the comment and point out how they address it. For my second concern, the authors need to put the real number of treated questions in each subsample.

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Response

We apologize for the typos and mistakes in our previous response.

The treatment group size was a typo in the letter. As reported in Section 4.2, the treatment group contains 1,766 questions; 3,613 is the total across both treatment and control groups. This was purely a reporting error in the letter; all statistical analyses were conducted on the correct samples.

The number "993" was also a reporting error due to a typo. The actual statistical analyses were conducted on correct subsamples, with a high-quality first answer subsample (top 25%): N = 1,697; and with a low-quality first answer subsample (bottom 25%): N = 1,945.

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Deleted: We reported the sample size in Online Appendix C ("Overview of Main Analyses: Sample Sizes and Selection Criteria") as shown below. In the revision, we further clarify these numbers in the main text of Section 6.3.2.3. We hope this fully addresses the reviewer's concerns.¶
Table C. Overview of Main Analyses: Sample Sizes and Selection Criteria¶
Table Number ... [3]

As stated in our opening remark, to make the paper more focused and to ensure the results align more strongly with our updated theoretical framework, we removed all netlike-related results in the manuscript. But we discussed the analysis details here in response to the reviewer's concern.

#R2-9

R2-9. I suggested the authors run a t-test for their randomization checks. The authors followed. Hence, I am good on that.

Response

We are happy you are satisfied with this response.

#R2-10 (5 mins)

R2-10. I questioned how the authors claimed that AI-generated answers appear within five minutes after question posting. They responded that they monitored/observed the platform in real time.

There are 1,766 questions in the treatment group, which contain an AI answer. Those 1,766 questions are posted across 150 days. I am really curious about how the authors monitor and observe. They should explain the process more clearly. Do they hire students to manually monitor/observe the platforms every five minutes or set up a script to automatically collect the platform every five minutes (does the platform allow them to do so?), or ask the platform to save a snapshot of the question page every five minutes? By saying monitoring/observing in real time without providing the procedures does not help.

Response

Sorry for the confusion. To clarify the underlying mechanism: on this platform, an AI-generated answer is, by design, always positioned before any human-generated answer for a treated question. The AI answer is not interleaved with human answers according to posting time; it is anchored to the top of the answer list.

Due to the time needed for AI answer generation, there is a small delay. In practice, this window is negligible relative to human response latency, since the first human answer is posted 22.067 hours on average. It will be faster than most humans.

The five-minute appearance of the AI answer was produced by us manually repeatedly monitoring the platform at the beginning of this research. What's more, it was produced by the platform's own procedure. When a new question was posted during the experimental period, the platform's system randomly assigned it to the treatment or control condition, and for treatment questions, the internal GenAI generated and posted its answer automatically after the question appeared.

The five-minute figure describes our manual observation cadence on our side. We have revised the relevant text to state the procedure explicitly and to attribute the timing to the platform's automated assignment.

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#R2-11

R2-11. The authors' explanation for the question's zero length makes sense.

Response

We are happy you are satisfied with this response.

#R2-12 (questioner random effect)

R2-12. In the earlier round, I suggest that the authors add a question poster/questioner-level fixed effect to remove questioner-level confounders.

In response, the authors clarified that the questioner characteristics are unlikely to confound the effect because treatment assignment is randomized at the question level. Adding questioner fixed effects reduces the sample size substantially, and excluding single-question posters may affect the generalizability. Conducting the question poster random effects and found consistent results.

I would respectfully disagree with the authors' arguments and want to ask for more clarification. First, treatment assignment is randomized, but we still need to test whether the randomization works. The authors run balance tests on observable characteristics and found no systematic differences, but those unobservable characteristics that can confound the results cannot be tested.

Second, adding questioner-fixed effects reduces the sample size substantially. The authors need to show the statistics of the samples in detail. I don't think focusing on questioners posting on more than one question will lose the generalizability. I may think the platforms want to know whether their way of using generative AI would affect both new and existing users (who post more than one question).

Third, I appreciate that the authors run the questioner random effect model. However, I may not think the key assumption for random effects models suits this context. To run the random effect model, the authors need to show E(u_i|X_i) = 0, while they can only get E(u_i|treatment) = 0 through the randomization. Therefore, the assumption of running the random effect does not hold.

Response

We thank the reviewer for the continued guidance on this point and for encouraging us to examine the identification more closely. We took the three concerns seriously and looked into each of them in turn.

On the sample restriction from questioner fixed effects. Implementing questioner fixed effects requires dropping all posters who contributed only a single question during our observation period, because such observations provide no within-questioner variation. In our data, 934 of the 3,613 questions were posted by single-question posters. Removing them would reduce the estimation sample by this amount and lower the statistical power available for our tests. We therefore report the questioner random effects specification, which retains the full sample,.

On whether the randomization worked. We agree that randomization should be verified rather

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than assumed. Two points are relevant here. First, the assignment was carried out by the platform (SegmentFault) as part of its own randomized experiment, so the randomized nature of the treatment is a feature of the design rather than an inference on our part. Second, we conducted a series of randomization checks on observable characteristics and did not find evidence of imbalance. We report these checks in the revised manuscript.

On unobservable characteristics that cannot be tested directly. The reviewer is correct that unobservable question-level characteristics cannot be measured directly, and this limitation is shared by any observational or experimental specification. What we can do is assess how sensitive our estimates would need to be to unobserved confounding before the conclusions would change. We therefore report the Oster (2019) bounds, and the results indicate that our estimates are robust to plausible degrees of unobserved omitted variable bias.

On the assumption underlying the random effects model. We would add one brief clarification. Because our parameter of interest is the treatment effect, the consistency of that coefficient depends on the error term being uncorrelated with treatment assignment, which the platform's randomization ensures. The additional covariates enter to improve efficiency rather than to achieve identification of the treatment effect. In this sense the condition we rely on for the coefficient of interest is met by the design.

We have updated the manuscript accordingly and describe these analyses in the revised results and robustness sections.

#R2-13

> *R2-13. I suggested running question-level aggregation when examining the impact on the answer viewer because the IV and DV are not at the same level. The authors followed and found consistent results. I am good with that.*
>
> *R2-14. I suggested the author justify why they used the cubic polynomial, and they followed. I am good with that.*
>
> *R2-15. I suggested the authors verify whether users see the AI-generated answers in the question list page. They followed and provided the Wayback Machine proof. I am good with that.*

Response

We are happy you are satisfied with these responses.

#*R2-14

> *R2-16. I raised a concern about the results of Table 5 in the last round. Specifically, Table 5 reveals that there is no statistical significance in the differences in the number of answers between the questions before and during the experimental period for the control questions. However, the different numbers disclosed by the authors reveal a big difference. For example, the average number of answers for control questions is 1.372, and the overall number of answers per question (including both the experimental period and pre-experimental period) is 1.9.*

The authors responded that the difference between 1.9 and 1.372 is not statistically significant and falls within normal sampling variation.

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I respectively disagree. First, the authors need to show how they conduct the statistical analysis. The overall number of answers per question is 1.9, while the experimental period is 1.372, signaling that the pre-experimental period has a number of answers per question higher than 1.9. The authors need to show the numbers and be more transparent.

Second, the authors attribute the difference to "normal sampling variation", without specifying the sampling process and the variance.

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Response

We thank the reviewer for the comment.

First, we want to clarify that the two numbers 1.372 and 1.9 come from **different populations** and cannot be directly compared to infer a trend. 1.372 is on the control group in the post-experiment period, and 1.9 is on the entire population. Our statement on *"normal sampling variation"* was intended to explain that the two numbers come from different **samples.**

Second, it should be noted that pre-experiment questions have been on the platform longer and have mechanically accumulated more answers over time. For the same reason, a raw comparison conflates time-on-platform with behavioral change. The fact that 1.9 is larger than 1.372 does not necessarily indicate a decrease in the number of answers.

Third, our studied period is small. Even though there is a decreasing trend, it may not be reflected or affect our analysis.

Due to these reasons, our spillover test compares **pre-experiment questions** (N = 4,137; mean = 1.482, SD = 1.131) against **control-group questions during the experiment** (N = 1,847; mean = 1.305, SD = 1.172). These two groups of questions are essentially the same type of questions without AI answers. Furthermore, we employ a **regression that controls for a cubic polynomial of question age** and other covariates. Once the mechanical age effect is removed, the coefficient on the post-AI deployment indicator is **−0.022 (SE = 0.015;** $p > 0.10$**)**. Our claim that *"the difference is not statistically significant"* referred to this result.

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This confirms that the introduction of internal GenAI did not change the patterns of human answering for control-group questions.

Following the reviewer's suggestion, we have clarified this in Section 6.2.3.

#R2-15

R2-17. In the earlier round, I challenged the authors' way of testing the content quality mechanisms by suggesting that they should first test that the AI answers are of higher quality and second test whether this higher quality AI-generated answers demotivate users from generating answers.

In response, the authors followed but argued the validity of their original approach, which only controlled the first answer quality by citing highly influential papers.

Since the authors argued, I want to respond and raise the authors' incorrect operations of only including the controls to test a mechanism.

First, by including controls, the authors didn't test whether AI answers are of higher quality, leaving their proposed theoretical mechanisms for higher AI answer quality in the air and untested (as supported by the authors' response: "By holding answer quality constant (controlling for it), we remove the portion of the AI effect that works through quality differences.")

Second, I agree with the authors that (β_1 - β_1') represents the portion of the effect that operates through answer quality. However, the authors did not test whether this difference was significant in their original submission. Therefore, the authors cannot conclude whether the reduced effect of AI answers on human answers stems from answer quality.

Accordingly, in this new approach, we still need to understand whether the effect difference before and after including the answer quality is significant or not (i.e., whether the difference of -0.153 and -0.113 in Table 9 is significant or not).

In addition, the authors only used answer length as the answer quality measure. It can be problematic. The internal GenAI in-built prompt can control the length of AI answers. The authors need to test other quality measures that can also mediate the effects.

Response

We thank the reviewer for pressing on both points, and we address them in turn.

On the significance of the coefficient difference between −0.153 and −0.113, and on control-based mediation more generally, we no longer use the control-based approach to establish mechanisms. As described in our opening remark and in our response to R2-1, every mechanism in the revised paper is now derived directly from theoretical argument under the dual-process framework and tested through predictions stated in advance, rather than through the change in a coefficient after adding a control. There is therefore no longer a coefficient-difference test of the kind the reviewer refers to, and no corresponding new result, because that inference strategy has been removed from the paper.

On the measurement of AI answer quality beyond length, we make three clarifications and additions. **First, we clarify that answer length measures coverage rather than overall quality.** In our original framing, we imprecisely characterized length as a quality proxy. We now position it specifically as a measure of answer coverage, which captures the breadth and thoroughness of comprehensiveness. This is conceptually distinct from the dimensions captured by our original LLM-based measures. As detailed in Appendix B, the LLM assessment focuses on objective quality dimensions such as accuracy, clarity, and relevance, which evaluate how well an answer addresses what it chooses to cover rather than how much ground it covers. This distinction explains the divergent mediation patterns. Potential human answerers are deterred primarily by the perceived high coverage of the AI answer, a surface cue that can be inferred from length at a glance, rather than by fine-grained quality, which becomes available only through effortful evaluation.

Second, we highlight the 3 validation tests we conduct for the LLM-based quality measure. Failed AI answers (for example, "I'm sorry, I can't provide a detailed answer") received significantly lower quality ratings than substantive answers ($p < 0.001$). Human answers accepted by questioners received significantly higher ratings than non-accepted answers ($p < 0.05$). Answers with above-median or above-mean net likes received significantly higher ratings than those below the median ($p < 0.001$). These results support the validity of the measures.

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Third, we examined what answer length actually captures. Length can reflect perceived quality, or it can reflect perceived coverage, meaning the impression that an answer spans many aspects of the problem. To separate these two readings, we added a second content variable, the number of bullet points in the answer. GenAI answers in this setting are typically well formatted and well structured, and an answer that enumerates several points tends to convey broad coverage, which motivates a human perception of comprehensiveness. If length were operating mainly through perceived coverage, its effect should track this coverage variable rather than a direct quality signal. Our results indicate that the role of length does operate largely through perceived coverage and not through perceived quality. We report this analysis in the mechanism part.

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#R2-16

R2-18. I raised concerns about the measurement of the "first-answer seeking tendency." The authors clarified that they measured the preference/tendency using observed choice rather than stated intentions. I think it makes sense and can mitigate my concern.

Response

We are happy you are satisfied with this response.

#*R2-17

R2-19. I challenged the AI's position effect by arguing that the reduction in the effect of AI answers on the number of human answers stems from the fact that those who tend to generate the first answer usually provide high-quality answers (because they prioritize quality), and then these high-quality human answers deter others from contributing.

In response, the authors conducted an additional analysis by including the hasHighQuality1Ans_i (=1 if the first human answer has an above-median LLM-generated content quality score) as the control variable and found that this variable positively affects the answerNum_i.

I feel my concern was not well addressed. A more direct test is to only focus on the control group and regress the answerNum_i on the hasHighQuality1Ans_i. If it has a positive effect, my early argument was disapproved.

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Response

We thank the reviewer for proposing this more direct test. We first report the analysis and then note the change in scope.

Following the suggestion, we restricted the sample to the control group and regressed answerNum on hasHighQuality1Ans. The estimated coefficient is statistically insignificant (See Table below). This result indicates that a high-quality first human answer does not deter subsequent human contributions.

Figure R2-17. Testing R2's Alternative Explanation That First-Position High Quality Answerer Deter Others' Participation

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	(1)
	DV: answerNum$_i$
hasHighQuality1Ans$_i$	0.012
	(0.015)
other controls	√
answer timing FE	√
conly control sample	√
N	1,451

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As the reviewer noted, this positive coefficient is inconsistent with the alternative explanation that first-position high quality answerer deter others' participation.

As stated in our opening remark, to make the paper more focused and to ensure the results align more strongly with our updated theoretical framework, the position effect has been removed from the manuscript in this revision.

We report the test above so that the reviewer can see the alternative explanation does not hold, and we note that the broader position argument is no longer part of the paper.

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#R2-18

2-20. I commented that the source effect was not tested.

The authors responded that "the source effect is tested in Section 6.3.1.3 Table 11." Specifically, "After controlling for the quality of the first answer (content effect) and whether the first human answerer has a first position seeking tendency (position effect), the estimated treatment effect becomes smaller but remains statistically significant. This residual effect, the portion of the AI impact that persists after accounting for content quality and position, indicates that the mere fact that an answer is provided by AI (rather than a human) independently affects user behavior."

I respectively disagree because there can be other reasons aside from the position and content, like anchoring reference, cognitive offloading, etc., for explaining the residual effect.

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Response

We agree with the reviewer that the earlier residual, defined as what remained after controlling for content and position, could reflect several forces and should not be labeled a single-source effect by elimination. Our revised framework addresses this directly because we test AI label effect and AI content directly based on a new theoretical framework.

The forces the reviewer names are now theorized inside the framework rather than left in a residual. Anchoring is now the explicit core of H4: the weak competitor judgment formed from the bare AI label acts as a kind of anchor on subsequent effort. Having judged the rival to be weak, the member expects that even a fairly modest answer will be enough to come out ahead. The law of least effort holds that people supply effort only up to the level a goal requires, and no further (Kahneman 2011). Cognitive offloading operates through the law of least effort that governs the fast stage. What we previously called the residual source effect is now positively theorized as the AI label effect, that is, the weak competitor perception that the bare label triggers through the availability heuristic in H1, rather than an unexplained remainder.

Consistent with this, and as described in our opening remark and our responses to R2-1 and R2-15, we no longer use the three-way control-based decomposition into position, content, and source. Each effect is now derived from theory and tested through a prediction stated in advance, so there is no residual term to interpret.

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#R2-19

R2-21. I commented that the experts should be domain-specific. The authors followed. I am good on that.

Response

We are happy you are satisfied with this response.

#R2-20 # R2-8

R2-22. I challenged the subsample sizes in the test of the expectation-based mechanism.

The authors responded that they explained in their response to my comment R2-8.

As a clarification to the R2-8, the authors didn't address my comments regarding this subsample size.

Response

We appreciate the suggestion of comment, as explained in R2-8, we apologize that our earlier reply to R2-8 did not directly report the subsample sizes for the expectation-based test. To be fully transparent, the earlier "993 versus 993" figures were a reporting error in the letter. The actual analyses used the high-quality first-answer subsample (top 25%, N = 1,697) and the low-quality first-answer subsample (bottom 25%, N = 1,945), as now documented in Online Appendix C (Table 13).

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We also note, as stated in our opening remark and our response to R2-6, that this revision removes all results using net likes as the dependent variable, which includes the expectation-based (elevated expectations) mechanism itself. We have reported the corrected subsample sizes here for completeness and to confirm that the original analyses were conducted on the correct samples.

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#R2-21

Here are some of the new comments I have while reading the revised paper. Since this is a revision, I focused only on comments that raised major concerns.

1. The reputation scores of the questioners are different across the treated and control groups, *where the reputation scores of the questioners in the control group are higher. This is concerning. The authors ran the balance tests using the log reputation score, which shows insignificant (p=0.065). How about the raw reputation score?*

Response

Thank you for raising this point.

The raw reputation score in our sample is highly right-skewed (Please see figure below, left panel), which severely violates the normality assumption underlying the t-test. The log transformation brings the distribution much closer to normality and better satisfies the statistical requirements (Please see figure below, right panel).

Following the suggestion of the reviewer, we conducted a Mann-Whitney-Wilcoxon test for balance test of the reputation score, which does not assume normality. The result (W = 1,667,892, p = 0.238) confirms that there is no statistically significant difference in questioner reputation between the treatment and control groups.

The revisions are reported in Online Appendix A in the paper.

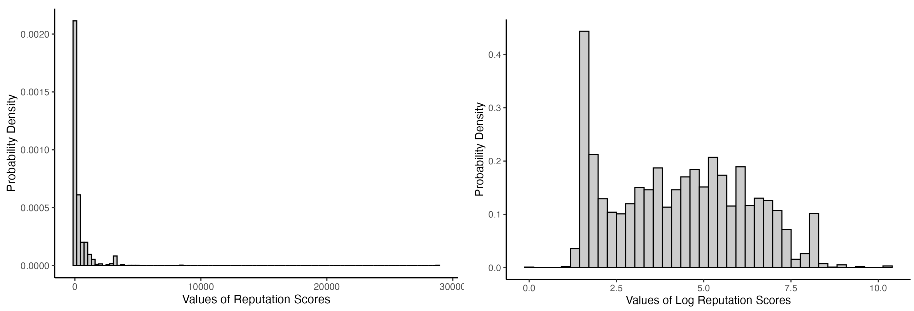

Figure R2-21. Distribution of The Reputation Scores

#*R2-22

2. The authors progressively add controls that look good, but they need to start with no controls (including removing the question timing FE)

Response

Following the suggestion of the reviewer, we have included a baseline specification with no control variables and no question timing fixed effects. The treatment effect remains consistent in both direction and statistical significance. This result is reported below and further confirms the robustness of our findings.

Table R2-22. The impact of AI-generated answers on the answer supply and viewer recognition

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)

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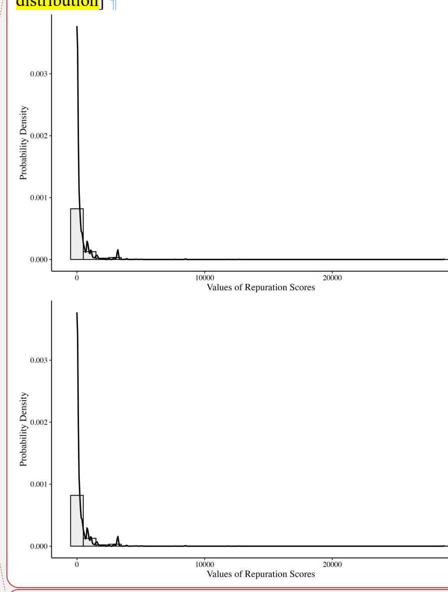

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Deleted: Thanks for the suggestion. We have conducted additional test on raw, reputation score the p value is 0.02130587. In the revision, we have reported this in the paper.

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	DV: log of the number of human answers provided within 7 days (answerNum$_i$)				DV: number of net likes received by an answer (netlikeNum$_{ij}$)			
AI_i	-0.052 ***	-0.050 ***	-0.048 ***	-0.046 ***	-0.049 **	-0.052 **	-0.052 **	-0.046 *
	(0.011)	(0.011)	(0.011)	(0.011)	(0.020)	(0.020)	(0.020)	(0.019)
question controls			√	√			√	√
questioner controls				√				√
question timing FE		√	√	√				
answer timing FE						√	√	√
N	3,613	3,613	3,613	3,613	4,670	4,670	4,670	4,670

As stated in our opening remark, to make the paper more focused and to ensure the results align more strongly with our updated theoretical framework, we have respectfully placed this analysis here but removed all netlike-related results in the manuscript.

#*R2-23

3. The netlike number has negative values. How did the authors adjust the non-negativity for running the Poisson model in column 6 of Table 5? If the authors can shift/adjust netlikeNum for the non-negativity, they can also do it to run a log-transformation model for netlikeNum.

Response

For the Poisson model, we shifted *netlikeNum* by adding a constant "3" (the minimum value of the *netlikeNum*) to satisfy the non-negativity requirement.

We also estimated a log-transformation model using log(*netlikeNum* + 3 + 1) as an additional robustness check. The results remain consistent across both specifications and further validates our findings. This analysis is reported below.

Viewer Recognition of human answers. Table R2-23 addresses the concern that answer timing fixed effects may be too restrictive, given that net likes accumulate differently over different time periods. We then adopt a more flexible approach to capture the non-linear relationship between question age and net likes accumulation. Column 1 presents the baseline specification without controlling for answer age, while Columns 2-4 progressively add polynomial controls: linear (Column 2), quadratic (Column 3), and cubic (Column 4) terms of answer age. Column 5 reports our preferred specification with answer timing fixed effects. Column 6 employs the logarithm of the net number of likes as the dependent variable. Column 7 implements a Poisson model. (Since the minimum number of likes in the data is -3. We adjust the analyses in columns 6 and 7 for non-positive values by shifting by 3.) All specifications show consistent negative effects

Table R2-23. Robustness checks: alternative model specifications for viewer recognition

	OLS						Poisson
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
DV:			*netlikeNum$_{ij}$*			log [*netlikeNum$_{ij}$* +shift + 1]	*netlikeNum$_{ij}$* +shifted
AI_i	-0.044 *	-0.044 *	-0.044 *	-0.044 *	-0.046 **	-0.010 **	-0.014 *

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	(0.020)	(0.020)	(0.020)	(0.020)	(0.019)	(0.004)	(0.006)
controlling for answer age (age_j)		√	√	√			
controlling for age_j^2			√	√			
controlling for age_j^3				√			
answer timing FE					√	√	√
other controls	√	√	√	√	√	√	√
N	4,670	4,670	4,670	4,670	4,670	4,670	4,670
log likelihood							-7408.622
pseudo R^2	-	-	-	-	-	-	-0.022

Notes: 1. ***p<0.001; **p<0.01; *p<0.05.
2. Answer timing FE indicates the date on which the answer was posted.

As stated in our opening remark, to make the paper more focused and to ensure the results align more strongly with our updated theoretical framework, we have respectfully placed this analysis here but removed all netlike-related results in the manuscript.

#*R2-24

4. Human answerers select to provide answers to questions without an AI answer because they want to gain recognition. The treatment effect is at the answerer level, who can select to join the treatment or control group. Testing the potential spillover effects is the right direction. However, more work is needed. Table 7 should follow Table 3, gradually adding controls, and all robustness checks conducted for the main effects should be applied to test for spillover effects.

Response

Following the suggestion of the reviewer, we have revised the spillover analysis to mirror the progressive control structure required in R2-22. We also applied all robustness checks conducted for the main effects to the spillover tests.

All results remain consistent across specifications, showing no significant spillover effects. The updated results are reported below.

Table R2-24-1. Testing Potential Spillover Effect

	(1)	(2)	(3)	(4)	(5)	(6)
	DV: log of the number of human answers provided within 7 days ($answerNum_i$)			DV: number of net likes received by an answer ($netlikeNum_{ij}$)		
post-AI deployment control group indicator	-0.027	-0.022	-0.022	-0.024	-0.018	-0.016
	(0.015)	(0.015)	(0.015)	(0.027)	(0.026)	(0.026)
question controls		√	√		√	√
questioner controls			√			√
controlling for a cubic polynomial of question/answer age	√	√	√	√	√	√
N	5,984	5,984	5,984	8,961	8,961	8,961

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Table R2-24-2. Robustness Checks For Spillover Effects: Alternative Model Specifications For Human Answer Supply

	OLS		Poisson	ZIP	ZINB
	(1)	(2)	(3)	(4)	(5)
	DV: log of the number of human answers provided within 7 days (*answerNum*$_i$)		DV: the number of human answers provided within 7 days		
post-AI deployment control group indicator	-0.022	-0.026	-0.019	-0.019	-0.019
	(0.015)	(0.015)	(0.042)	(0.042)	(0.042)
other controls	√	√	√	√	√
controlling for a cubic polynomial of question age	√	√	√	√	√
questioner random effect		√			
N	5,984	5,984	5,984	5,984	5,984
AIC		2691.088		17086.635	17086.635
BIC		2979.053			
log likelihood		-1302.544	-8501.317	-8501.317	-8501.317
deviance			5177.952		
pseudo R^2			0.019		

Table R2-24-3. Robustness Checks: Alternative Model Specifications For Viewer Recognition

	OLS		Poisson
	(1)	(2)	(3)
DV:	*netlikeNum*$_{ij}$	log [*netlikeNum*$_{ij}$ (shifted) + 1]	*netlikeNum*$_{ij}$ (shifted)
AI$_i$	-0.016	-0.003	-0.005
	(0.026)	(0.006)	(0.022)
other controls	√	√	√
controlling for a cubic polynomial of answer age	√	√	√
N	8,961	8,961	8,961
log likelihood			-14301.496
pseudo R^2			-0.001

As stated in our opening remark, to make the paper more focused and to ensure the results align more strongly with our updated theoretical framework, we have respectfully placed this analysis here but removed all netlike-related results in the manuscript.

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#R2-25

5. There is misalignment between the arguments and the empirical test. Arguments: When AI answers appear immediately in the first position, they eliminate this opportunity for humans, reducing potential answerers' incentive to contribute. Empirical test: Our mediator variable, hasFirstPositionSeeker1Ansi, equals one if the first human answer came from such a user. The mediator captures a key prediction of the position effect mechanism: if AI deters participation by occupying the first position, it should particularly discourage "first position seekers" from contributing, making them less likely to provide the first human answer. Not the first human answer, but the overall contribution.

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Response

Following the suggestion of the reviewer, we constructed a new mediator variable that counts the number of answerers provided by "first-position seekers" to each question. As shown in Column 2 of the table below, AI significantly reduces this count. When we include this mediator in the main regression, the AI coefficient decreases by a larger magnitude compared to the previous specification, showing that "first-position seekers" significantly reduced their contribution which caused the overall submission reduction.

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We show these robustness results in Table R2-25.

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Table R2-25. Additional Evidence for Position Effect

	(1)	(2)	(3)
DV:	$answerNum_i$	$firstPositionSeekerNum_i$	$answerNum_i$
AI_i	-0.153 ***	-0.261 ***	-0.086 ***
	(0.010)	(0.027)	(0.008)
$firstPositionSeekerNum_i$			0.257 ***
			(0.005)
question timing FE	√	√	√
other controls	√	√	√
N	3,235	3,235	3,235

As stated in our opening remark, to make the paper more focused and to ensure the results align more strongly with our updated theoretical framework, the position effect has been removed from the manuscript in this revision. Because the position effect is no longer part of the paper, we do not further develop the argument about the position effect.

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#R2-26

6. Likes and comments require different levels of cognitive effort, which is not purely explained by attention-only. There are no significant results associated with commenting activity, but this does not mean there is no attention diversion.

Response

Thank you for this thoughtful comment. We agree that likes and comments involve different levels of cognitive effort. However, our argument does not rest on equating these two behaviors.

Rather, we reason that both behaviors share a necessary precondition: the viewer must read and attend to the human answer. If attention diversion were the primary driver, we would expect a detectable reduction in commenting as well, because fewer viewers would encounter and read human answers in the first place. The absence of any reduction in comments suggests that viewers continue to engage with human answers at a similar level. Combined with our finding that the negative effect on likes concentrates among questions where the AI first answer is of particularly high quality (top 25%), the overall pattern is more consistent with elevated evaluative standards than with attention diversion.

As stated in our opening remark, to make the paper more focused and to ensure the results align more strongly with our updated theoretical framework, we removed all netlike-related results in the manuscript. But we retain the underlying logic here in response to the reviewer's concern.

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Reviewer 3 Comments

#*R3-0

I thank the authors for resubmitting the manuscript. I was Reviewer 3 in the previous round. The authors have substantially revised the paper by providing clearer information about the business context, strengthening the empirical analysis, positioning the work more effectively within the literature, and offering additional evidence on the underlying mechanisms. Several of the concerns I raised previously have been addressed, including the validity of the randomized field experiment and the distinction between this study and related work. However, I still have a number of questions that I believe the authors should clarify, as outlined below.

Response

We sincerely thank Reviewer 3 for acknowledging the substantial revisions we made to the manuscript, including the clearer business context, strengthened empirical analysis, improved positioning within the literature, and additional mechanism evidence. We appreciate the recognition that several previously raised concerns have been addressed, particularly regarding the validity of our randomized field experiment and differentiation from related work. The reviewer's continued engagement with our work provides valuable guidance for further improvements. We will carefully address the remaining questions outlined in the review.

In this round of revision, we benefited greatly from extensive communication with the **Senior Editor, including both email correspondence and offline face-to-face discussion**. Following SE's **detailed and actionable guidance**, we have made the following major changes. **First**, we substantially refined the theoretical framework along the competition direction suggested by the Senior Editor, establishing the theoretical hypotheses under a coherent and overarching logic. **Second**, we realigned our empirical results with this framework, which involved both adding results that speak more directly to the theory and removing those that were less central to it. **Third**, we sharpened the positioning of our contribution by more explicitly articulating how our study differs from prior work. We elaborate on each of these changes in our point-by-point responses below.

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#R3-1 (content effect)

1. Mechanism

In the previous round, I expressed concern that many of the empirical tests did not provide direct evidence in support of the proposed mechanism. I continue to have this concern, as the authors have not substantially revised or extended their analyses in the mechanism exploration part.

a. Content effect

The authors show that the impact of AI on the number of answers is mediated by the length of the first answer, but not by other quality measures generated by the LLM. I do not find this analysis to be strong evidence for a content-based mechanism. First, answer length is not, in itself, a reliable or robust measure of content quality. If the authors are unable to obtain consistent results using other LLM-generated quality measures, does this imply that the *content-effect mechanism is not supported, or at least only weakly supported? Alternatively, does this raise questions about the validity and reliability of the LLM-based quality measures themselves? In addition, it would be helpful if the authors could also report results from a formal causal mediation analysis for this mechanism.*

Response

We thank the reviewer for pressing on the mechanism evidence. Before addressing the specific points, we describe two changes that run across our responses. Later comments refer back to this note rather than repeating it.

First, a reorganization of how the theory is presented. We have rebuilt the theoretical development based on the email communication and offline face-to-face instruction with SE. It begins from: **1) a competitive encounter perspective** motivated by competition dynamics theory (Chen et al., 2007; Chen, 2009), which explains how the member appraises the AI rival; and **2) a single overarching account Kahneman's fast-and-slow theory** (Evans & Stanovich, 2013; Kahneman, 2003, 2011), which explains which signal can affect each decision and when that signal becomes available..

Grounded in new framework, we now claim that the effect of an AI answer depends on the **AI profile**. The AI label alone signals that a competitor has entered the question. In a community that already views AI-generated content with suspicion, users treat this competitor as a weak one, and the presence of a weak competitor motivates human contribution (H1). The AI length works in the opposite direction. When the AI answer is long and appears comprehensive, potential answerers judge that little of value remains to add, and their contribution falls (H2). The positive prediction is thus derived from theory before analysis and is stated in the hypotheses themselves, not inferred after the fact.

Our results confirm this framework. The AI label alone effect on supply is positive and significant (0.188, $p<0.05$), the interaction with AI answer length is negative (-0.046, $p<0.001$), and the average total effect turns negative only because AI answers in our setting carry substantial content, which is identically the total deterrence effect in the original manuscript.

Second, a deliberate narrowing of scope to keep the story focused. The removal is a choice about narrative focus and length, not a retraction of the earlier logic. We removed three elements from the main text. (1) All results that use net likes as the dependent variable, together with every recognition mechanism built on them, are removed. (2) The position effect is removed from the main text. (3) In the response letter, we still engage fully with the reviewer's earlier concerns about these removed elements, in order to show that the original reasoning was sound.

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We now respond to the three concerns about the content effect.

First, we clarify that answer length measures coverage rather than overall quality. In our original framing, we imprecisely characterized length as a quality proxy. We now position it specifically as a measure of answer coverage, which captures the breadth and thoroughness of comprehensiveness. This is conceptually distinct from the dimensions captured by our original LLM-based measures. As detailed in Appendix B, the LLM assessment focuses on objective quality dimensions such as accuracy, clarity, and relevance, which evaluate how well an answer addresses what it chooses to cover rather than how much ground it covers. This distinction explains the divergent mediation patterns. Potential human answerers are deterred primarily by the perceived high coverage of the AI answer, a surface cue that can be inferred from length at a glance, rather than by fine-grained quality, which becomes available only through effortful evaluation.

Second, we highlight the 3 validation tests we conduct for the LLM-based quality measure. Failed AI answers (for example, "I'm sorry, I can't provide a detailed answer") received significantly lower quality ratings than substantive answers ($p < 0.001$). Human answers accepted by questioners received significantly higher ratings than non-accepted answers ($p < 0.05$). Answers with above-median or above-mean net likes received significantly higher ratings than those below the median ($p < 0.001$). These results support the validity of the measures.

Third, we now provide detailed causal mediation results required for the original results in Table R3-1. We understand that the "formal causal mediation analysis" refers to the approach developed by Imai, Keele, and Tingley (2010). We did conduct this analysis in our previous revision, but we acknowledge that the presentation was not sufficiently detailed. In this revision, we report the full causal mediation results in a dedicated table (Table 12), with the average causal mediation effect (ACME), average direct effect (ADE), total effect, and the proportion mediated. The results confirm that the content channel accounts for approximately 26% of the total treatment effect. To keep the main text focused, we have removed the position effect and now new tests are not based on mediation analysis. So we report this table in the response letter rather than in the manuscript.

Table R3-1. Causal Mediation Analysis Results

	Content Effect	**Position Effect**
Effect Decomposition		
ACME	**-0.041*** **	**-0.031*** **
	[-0.063, -0.018]	**[-0.037, -0.024]**
Direct Effect (ADE)	-0.115***	-0.125***
	[-0.144, -0.083]	[-0.144, -0.104]
Total Effect	-0.156***	-0.156***
	[-0.175, -0.136]	[-0.176, -0.136]
Proportion Mediated		
% of Total Effect Mediated	**26.50%**	**19.70%**
	[23.6%, 30.4%]	**[17.5%, 22.6%]**

#R3-2 (position effect)

b. Position effect

I also find the analysis of the position effect confusing. The authors examine whether AI reduces the likelihood that the first human answer is provided by a "first-position seeker."

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However, it is not clear why the focus is specifically on the first human answer. It is also unclear how a user's prior tendency to post the first answer to a question should influence their decision to provide the first human answer (which is, in fact, the second answer overall when an AI answer is present). In addition, even if the presence of an AI answer reduces the number of users who seek to provide the first answer in other questions, this does not necessarily imply that the focal question will receive fewer answers overall; there may be sufficient users who are indifferent to answer position and still choose to contribute. Overall, I am not sure how this analysis provides compelling evidence in support of the proposed position-based mechanism.

Response

We thank the reviewer for these points, and we agree with them. We did assume that first-position seekers care about being absolutely first rather than first among human contributors. This assumption follows from our competitive framing. Because the human answerer and the AI compete for the same scarce resource, the limited attention and recognition of viewers, the AI answer is a rival for the first position and not merely background content. From this standpoint, an existing AI answer reduces the value of the first human answer for a first-position seeker.

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As stated in R3-1, to make the paper more focused and to ensure the results align more strongly with our updated theoretical framework, the position effect has been removed from the manuscript in this revision.

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#R3-3 (source effect)

c. Source effect

The authors argue that any residual effect of AI on the number of human answers, after controlling for both content and position effects, can be interpreted as evidence of a source effect. This inference is not well justified for at least two reasons. First, there may be alternative explanations for why the introduction of AI reduces the number of human answers that are not captured by the current controls. Second, any remaining residual effect could simply reflect measurement error or incomplete operationalization of the content and position mechanisms. Indeed, the earlier analysis of the content effect already suggests that the proposed quality measures are not ideal, which further undermines the claim that the residual effect can be cleanly attributed to a source-based mechanism.

Response

We agree with the reviewer. The residual-based inference cannot be defended, for the two reasons the reviewer gives. Alternative explanations that the current controls do not capture may drive the reduction in human answers, and any remaining residual may reflect measurement error or an incomplete operationalization of the other channels. We no longer rest any claim on a residual.

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In this revision we changed how the label enters the analysis. Previously we started from a total effect and then asked what explained the part that the content and position channels did not account for, and we labeled that remainder a source effect. We now claim that the effect of an AI answer depends on the **AI profile**. The AI label alone signals that a competitor has entered the question. In a community that already views AI-generated content with suspicion, users treat this competitor as a weak one, and the presence of a weak competitor motivates human contribution (H1). The AI length works in the opposite direction. When the AI answer is long and appears comprehensive, potential answerers judge that little of value remains to add, and their contribution falls (H2). The positive prediction is thus derived from theory before analysis and is stated in the hypotheses themselves, not inferred after the fact.

Our results confirm this framework. The AI label alone effect on supply is positive and significant (0.188, $p<0.05$), the interaction with AI answer length is negative (-0.046, $p<0.001$), and the average total effect turns negative only because AI answers in our setting carry substantial content, which is identically the total deterrence effect in the original manuscript.

We hope you find this way of theoretical development under a coherent overarching theoretical framework clearer and satisfactory.

#R3-4 (attention diversion)

d. Attention diversion mechanism

Although the effect of AI on the number of comments is statistically insignificant, the magnitude of the estimated coefficient is very similar to that for net likes. It may be hard to conclude that users' attention has not been diverted from human to AI answers.

Response

We thank the reviewer, and we clarify the logic of this test. The two outcomes are not comparable in the way that the similar coefficient magnitudes suggest, for two reasons.

First, they are measured on different scales. The net likes variable is not logged, so its coefficient reflects an absolute change, whereas the number of comments is logged, so its coefficient reflects an approximate proportional change. A similar coefficient magnitude therefore does not indicate a similar effect.

Second, the two outcomes capture different things. Net likes reflect viewer recognition of an answer, while comments reflect a more neutral activity that is driven by attention. Our test logic follows from this distinction. If attention were diverted from human answers to the AI answer, then both recognition and attention-driven activity should fall together, so the number of comments should decline as well. We do not observe a decline in comments. The pattern is therefore consistent with a reduction in recognition rather than a diversion of attention. For compactness, the analysis of net likes has been removed from the main text in this revision, but we retain the underlying logic here in response to the reviewer's concern.

As stated in our opening remark, to make the paper more focused and to ensure the results align more strongly with our updated theoretical framework, we removed all netlike-related results in the manuscript. But we retain the underlying logic here in response to the reviewer's concern.

#R3-5 (quality degradation)

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Formatted: Font: (Asian) Times New Roman *e. Quality degradation mechanism*

I believe the authors' analysis only shows that users who post answers to questions with AI involvement have a similar level of expertise as those who respond to questions without AI. This does not, however, rule out the possibility that the same type of users exert less effort on questions where an AI answer is present, which could result in lower-quality human answers.

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Response

We thank the reviewer for this comment, and we examined both pathways through which quality degradation could reduce the recognition of human answers. The main effect at issue is the decline in net likes on human answers when an AI answer is present. One explanation for this decline is quality degradation, in which human answers are of lower quality. Quality degradation could arise through two pathways. The first is a compositional shift, in which lower-expertise users replace higher-expertise users. The second is reduced effort, in which the same users invest less effort and produce lower-quality answers. In our understanding, the reviewer's point that the same type of users exert less effort on questions where an AI answer is present is the second pathway. **Effort is the channel through which the same users would produce lower-quality answers.** We tested both pathways.

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1. **We find no evidence of a compositional shift.** We measure expertise using each answerer's prior track record, specifically their count of previously accepted answers within the question's tag area. This metric reflects demonstrated domain competence. If AI discouraged expert participation, we would observe lower average expertise among human answerers on treated questions. Original Table 12, Column 2 shows no such reduction ($\beta = 0.067$, $p = 0.241$). The coefficient is positive but not significant, so expert answerers remain at least as willing to participate when AI answers are present.
2. **We find no evidence of reduced answer quality.** We regressed LLM-generated quality ratings of human answers on the AI treatment indicator. Table 12, Column 3 shows no association between AI presence and quality ratings ($\beta = -0.002$, $p = 0.975$). As a robustness check, we used answer length as an alternative quality proxy, since longer answers tend to be more comprehensive. Original Table 12, Column 4 shows human answers are slightly longer when AI is present, though the difference is not significant ($\beta = 0.076$, $p = 0.072$).

These results provide indirect evidence on the effort channel. Conditional on the composition of answerers being unchanged, if the same type of users were to invest less effort in the presence of an AI answer, the quality of their answers would be expected to decline, because quality is determined by effort. We do not observe a decline in quality. This suggests that effort is likely not reduced. We acknowledge that we do not have a direct measure of the effort an answerer invests in a given answer, so this evidence is indirect and is not a direct test of the effort channel. We are grateful to the reviewer for raising this point.

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As stated in our opening remark, to make the paper more focused and to ensure the results align more strongly with our updated theoretical framework, we removed all netlike-related results in the manuscript. But we retain the underlying logic here in response to the reviewer's concern.

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#R3-6 (elevated expectation)

f. Elevated expectation mechanism

I suggest that the authors consider using AI answer quality as a moderating variable to test this mechanism, as doing so may offer a more general and informative assessment of the underlying relationship.

Response

We appreciate this thoughtful suggestion. However, directly using AI answer quality as a moderating variable is not feasible in our experimental design: questions in the control group never receive an AI answer, and thus no AI answer quality measure exists for these observations, precluding a valid cross-group interaction test.

To address the underlying intuition that higher-quality AI responses should elevate viewer expectations more strongly, we adopt a related but implementable approach. We use the quality of the first answer (regardless of whether it is AI-generated or human-generated) as a continuous moderator. Importantly, we exclude observations where the first answer in the control group is provided by a human answerer, because in the control group AI treatment equals zero by definition; interacting a zero-valued treatment indicator with one's own quality to predict one's own outcome lacks clear theoretical interpretation.

With this refined sample, we estimate: $netlikeNum = \beta_0 + \beta_1\, AI \times quality1Ans + answer\ timing\ FE + X + \varepsilon$. The interaction coefficient is negative and statistically significant ($\beta_1 = -0.037$, $p < 0.05$), indicating that when the first answer is of higher quality, AI treatment leads to a larger reduction in net likes on subsequent human answers. This result provides direct support for the elevated expectations mechanism: high-quality initial answers (predominantly AI in the treatment group) raise viewers' evaluative standards, leading to harsher assessments of subsequent human contributions.

Table R3-6. Additional Test for Elevated Expectation Mechanism

	(1)
	DV: number of net likes received by an answer ($netlikeNum_{ij}$)
AI_i	0.169
	(0.103)
AI_i*$quality1Ans_{ij}$	-0.037 *
	(0.015)
other controls	√
answer timing FE	√
N	3,189

As stated in our opening remark, to make the paper more focused and to ensure the results align more strongly with our updated theoretical framework, we have respectfully discussed this analysis here but removed all netlike-related results in the manuscript.

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#R3-7 (theory)

2. Theory

In Section 3, the authors seek to theorize how internal GenAI may influence users' contributions to answers. However, there does not appear to be an overarching theoretical framework that explains why all three effects should shape contribution decisions. At present, each effect is discussed in isolation, which makes the overall theoretical argument appear somewhat ad hoc.

Response

We have rebuilt the theoretical development based on the email communication and off-line face-to-face instruction with SE. It begins from: **1) a competitive encounter perspective** motivated by competition dynamics theory (Chen et al., 2007; Chen, 2009), which explains how the member appraises the AI rival; and **2) a single overarching account Kahneman's fast-and-slow theory** (Evans & Stanovich, 2013; Kahneman, 2003, 2011), which explains which signal can affect each decision and when that signal becomes available..

The revised framework derives six predictions from three signals across the two decisions. Specifically, we specify which mode of processing governs each decision and state the condition under which it does so. The answerer's encounter unfolds in two stages. The first decision, whether to contribute at all, is binary and fast, and it is governed by System 1 because the only available information consists of surface cues, the AI label, and the AI answer length, which can be processed at a glance and cost almost nothing to process under the law of least effort. The second decision, how much effort to invest, is governed by System 2, and we state the condition for its engagement explicitly. Substantive answer quality is a slow signal that only deliberate reading can disclose, and System 2 engages because the competitive stakes, including a threat to the answerer's professional identity, are judged sufficiently important to override the law of least effort.

Each route assignment is tied to a specific, testable hypothesis. For answer supply, the AI label is a fast source cue and predicts a positive level effect in H1, length is a glance-based coverage cue and predicts a negative slope in H2, and quality requires reading and predicts no appreciable slope in H3. For first human answer quality, the label predicts a negative level effect in H4, length predicts no appreciable slope in H5, and AI quality sets a benchmark and predicts a positive slope in H6. The mappings are therefore stated in advance and examined empirically rather than asserted post hoc. We hope the revised theoretical development section makes the theoretical grounding clearer.

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#R3-8 (5 mins)

3. Other minor issues

a. On page 18, the authors state that "we observe that AI answers appear within five minutes of question posting." Could the authors verify whether AI answers consistently arrive faster than human answers?

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Response

Sorry for the confusion. To clarify the underlying mechanism: on this platform, an AI-generated

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Due to the time needed for AI answer generation, there is a small delay. In practice, this window is negligible relative to human response latency, since the first human answer is posted 22.067 hours on average. It will be faster than most humans.

#*R3-9

> *b. The authors state that they collected all questions and answers posted between September 2023 and February 2024. However, the analyses presented later in the paper also make use of pre-treatment data. The authors may want to correct this statement.*

Response

Sorry for the confusion. Our main experimental analyses focus on the 150-day experimental window from September 2023 to February 2024. However, for specific analyses such as the spillover effect analysis that compares user behavior before and after the experiment began, we utilized pre-treatment data to establish baseline patterns.

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We have revised the data collection description in the data collection section to accurately reflect both the data used in the paper.

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 # R3-10

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> *c. What is the distribution of the variable netlikenum? Does it exhibit a skewed distribution?*

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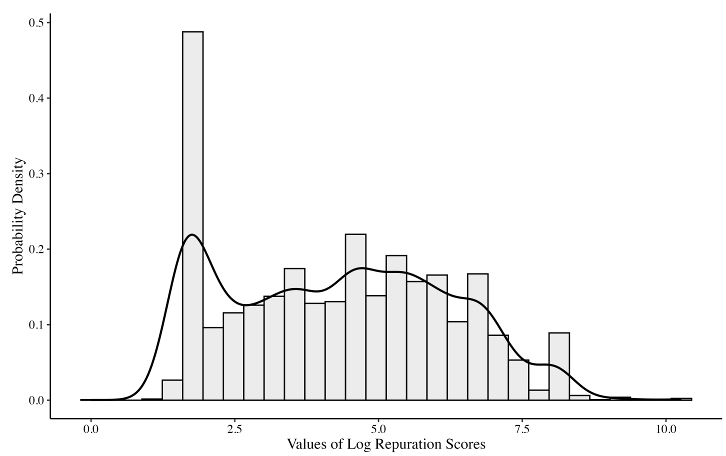
Response

Thank you for this question. The distribution of netlikeNum is indeed highly skewed (as shown below), with a mean of 0.288, and a standard deviation of 0.671.



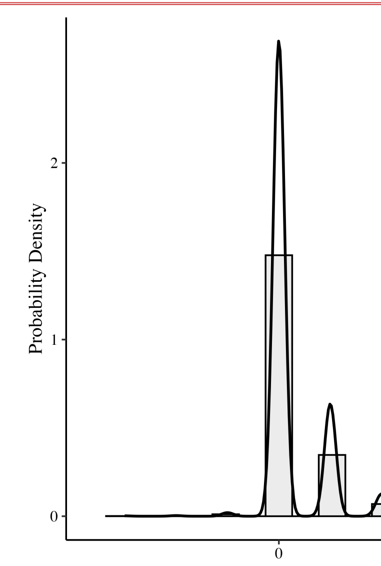


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Figure R3-10. Distribution of The netlikeNum

To tackle this distribution skewness, we estimate the model with both a Poisson model and a log-transformation model. The results are consistent across all specifications (See Table R3-10 below).

Viewer Recognition of human answers. Table R3-10 addresses the concern that answer timing fixed effects may be too restrictive, given that net likes accumulate differently over different time periods. We then adopt a more flexible approach to capture the non-linear relationship between question age and net likes accumulation. Column 1 presents the baseline specification without controlling for answer age, while Columns 2-4 progressively add polynomial controls: linear (Column 2), quadratic (Column 3), and cubic (Column 4) terms of answer age. Column 5 reports our preferred specification with answer timing fixed effects. Column 6 employs the logarithm of the net number of likes as the dependent variable. Column 7 implements a Poisson model. (Since the minimum number of likes in the data is -3. We adjust the analyses in columns 6 and 7 for non-positive values by shifting by 3.) All specifications show consistent negative effects

Table R3-10. Robustness checks: alternative model specifications for viewer recognition

	OLS						Poisson
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
DV:			*netlikeNum$_{ij}$*			log [*netlikeNum$_{ij}$* +shift + 1]	*netlikeNum$_{ij}$* +shifted
AI$_i$	-0.044 *	-0.044 *	-0.044 *	-0.044 *	-0.046 **	-0.010 **	-0.014 *
	(0.020)	(0.020)	(0.020)	(0.020)	(0.019)	(0.004)	(0.006)
controlling for answer age (*age$_j$*)		√	√	√			
controlling for *age$_j^2$*			√	√			
controlling for *age$_j^3$*				√			
answer timing FE					√	√	√
other controls	√	√	√	√	√	√	√
N	4,670	4,670	4,670	4,670	4,670	4,670	4,670
log likelihood							-7408.622
pseudo R^2	-	-	-	-	-	-	-0.022

Notes: 1. ***p<0.001; **p<0.01; *p<0.05.
2. Answer timing FE indicates the date on which the answer was posted.

As stated in our opening remark, to make the paper more focused and to ensure the results align more strongly with our updated theoretical framework, we have respectfully placed this analysis here but removed all netlike-related results in the manuscript.

#*R3-11

d. In the analysis presented in Section 7.2, should the authors account for the content similarity between the AI-generated answer and all human-generated answers?

Response

We thank the reviewer for this thoughtful suggestion.

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The analysis in original Section 7.2 aims to examine whether human answerers adjust their contributions to distinguish themselves from AI content. We only compare the first answer and the second answer to avoid the confounding effect of other prior answers from the human answerer

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Specifically, we measure the text similarity between the first answer and the second answer in two groups:

- In the treatment group, the first answer is AI-generated and the second answer is human-generated. The similarity score reflects how much the human answerer differentiates from the AI answer.
- In the control group, both the first and second answers are human-generated. The similarity score reflects the baseline level of differentiation between two human answers.

The reviewer's suggestion would measure the difference between AI-generated answers and human-generated answers, which answers an interesting but different research question. It does not allow us to test whether humans strategically adjust their answers in response to the appearance of a prior AI answer since there won't be a control group.

Deleted: Our finding shows that similarity is significantly lower in the treatment group (Columns 1-2 of Table 15). This result demonstrates that human answerers differentiate their responses when an AI answer is present, compared to when they respond to another human answer. ¶

We hope this clarification addresses the reviewer's concern.

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